

Nucleotide Metabolism: Synthesis & Degradation

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Objectives:

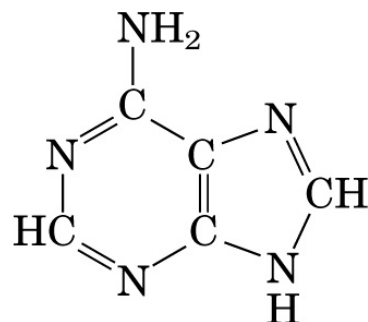
At the end of this lecture, it is expected to;

- State the Biological importance of Nucleotides
- Describe Nucleotides metabolism
 - Denovo synthesis of Purines nucleotides.
 - Salvage Pathways of Purine Synthesis
 - Degradation of Purines nucleotides.
 - Synthesis and degradation of Pyrimidine nucleotides.
 - Synthesis of deoxyribonucleotides.
- Explain the disorders related to Purine and Pyrimidine nucleotide metabolisms.

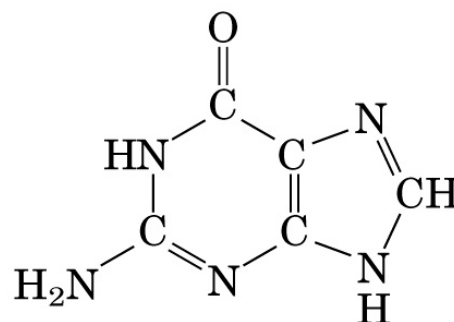
Biological Importance of Nucleotides:

- I. NTPs and dNTPs are activated precursors of RNA and DNA synthesis
- II. ATP - the universal currency of energy.
- III. ADP - the component of many coenzymes such as NAD⁺, NADP⁺, FAD, coenzyme A, etc.
- III. NTPs required in the activation of molecules in synthetic pathways.
 - GDP & GTP □ energy source for protein synthesis and other metabolic reactions
 - UDP - take part in galactose metabolism, glycogenesis and detoxification processes.
 - CDP & CTP - required for the biosynthesis of some phospholipids sphingomyelin
- IV. NTPs regulate protein function.
 - Through phosphorylation
 - Through binding and hydrolysis (G-proteins, Chaperones)
- V. Signal Transduction - formation of cyclic nucleotides
 - cAMP - a second messenger for many hormones, e.g. epinephrine, glucagon, etc.
 - cGMP - an intracellular signal or second messenger that can act antagonistically to cAMP.

Nitrogenous Bases: Purines & Pyrimidines

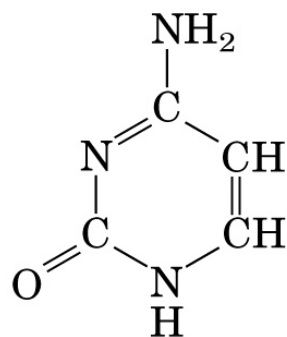


Adenine

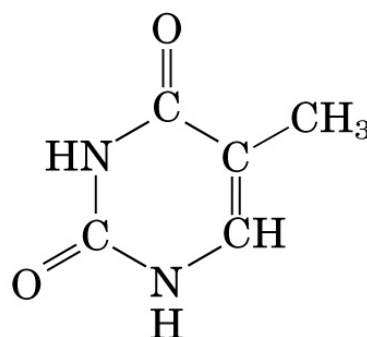


Guanine

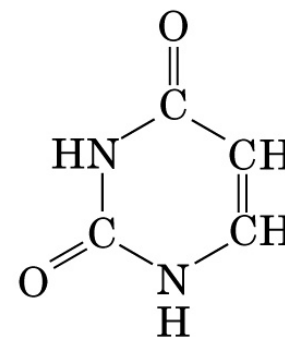
Purines



Cytosine



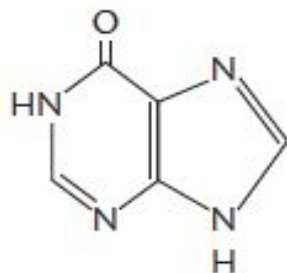
Thymine
(DNA)



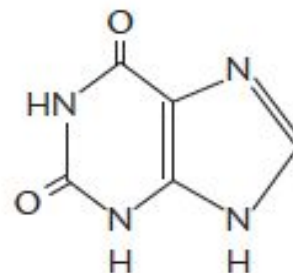
Uracil
(RNA)

Pyrimidines

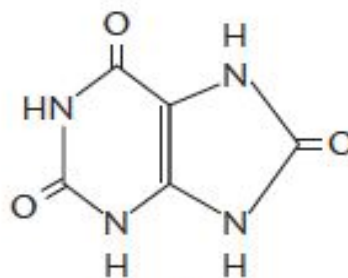
Intracellular minor purine bases



Hypoxanthine
(6-oxopurine)



Xanthine
(2,6-dioxopurine)

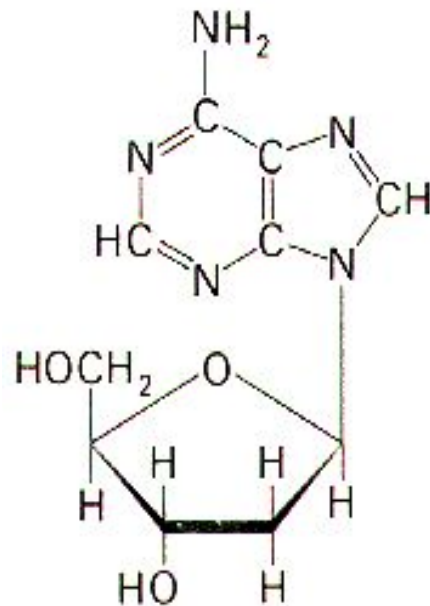


Uric acid
(2,6,8-trioxopurine)

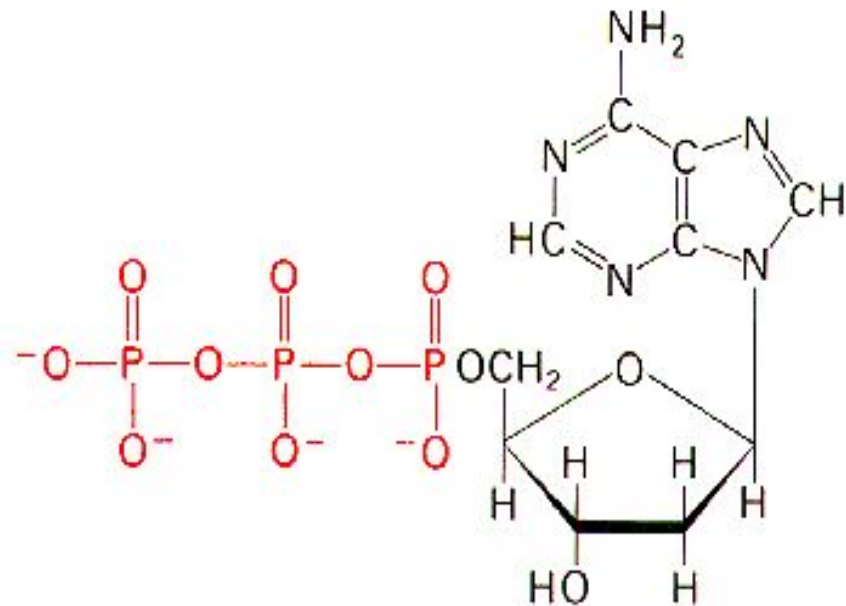
Nucleosides & Nucleotides

Nucleoside = Nitrogenous base – ribose

Nucleotide = Nitrogenous base – ribose – phosphate

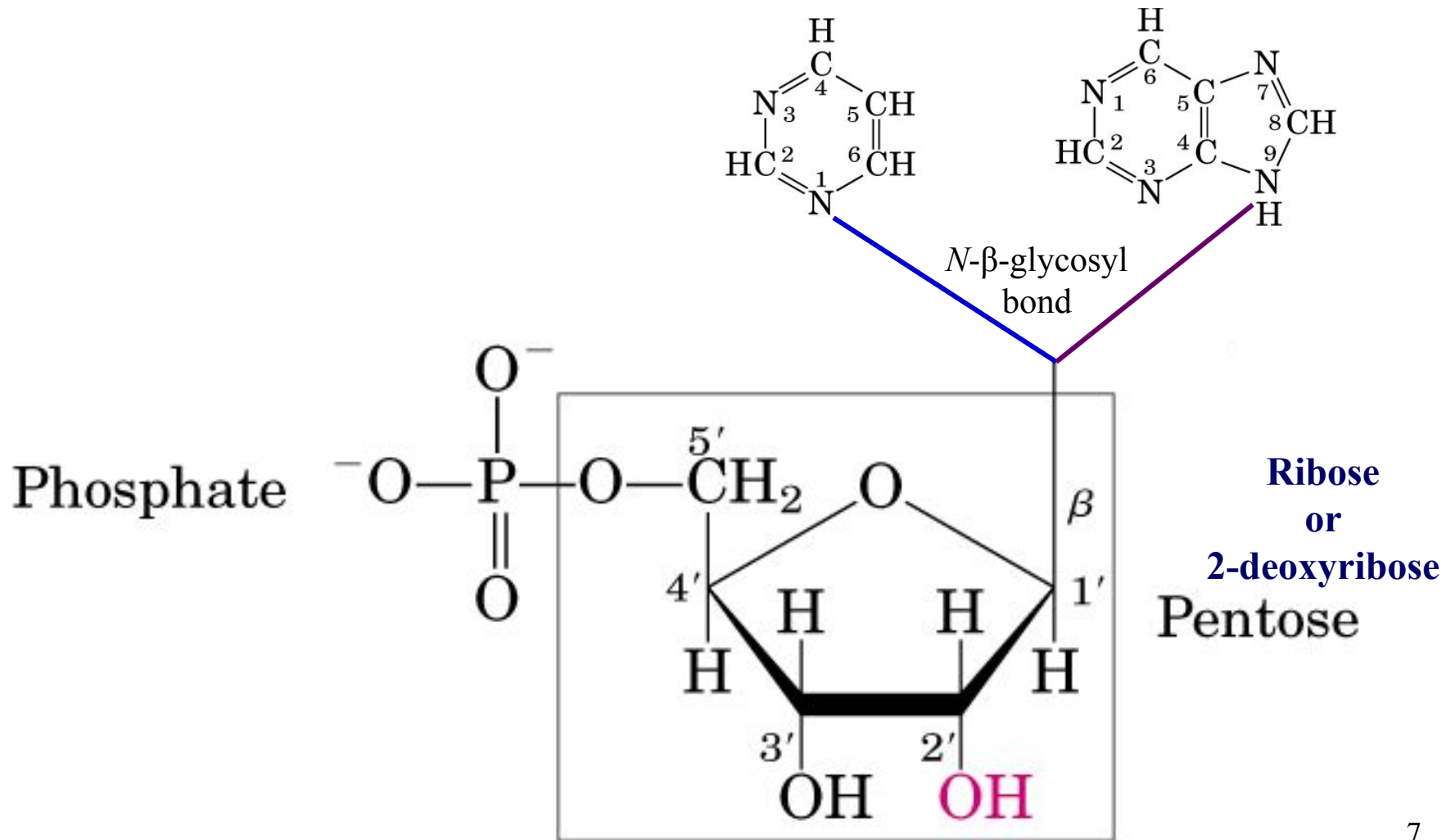


Deoxyadenosine
(A nucleoside)

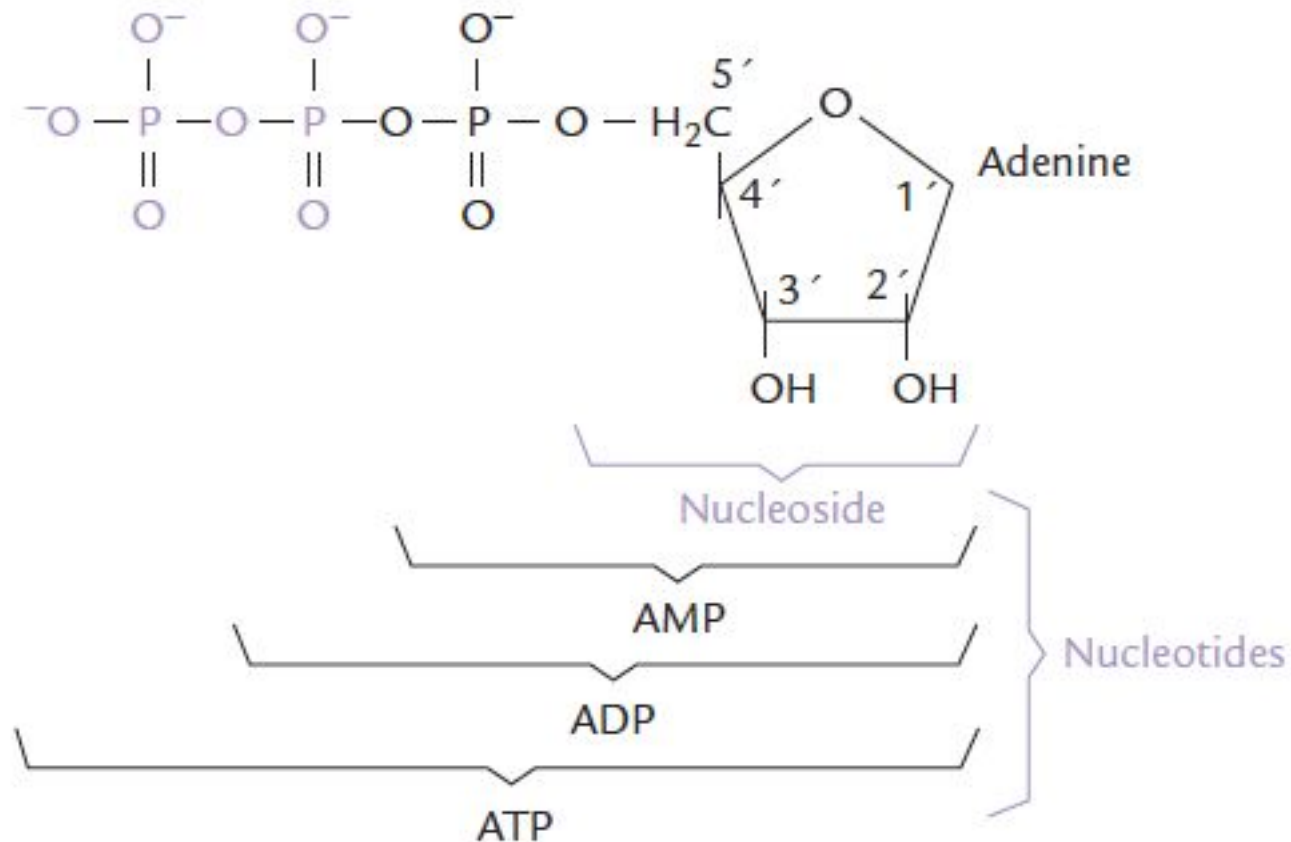


Deoxyadenosine 5'-triphosphate
(dATP)
(A nucleotide)

Structure of nucleotides



Structure of nucleotides

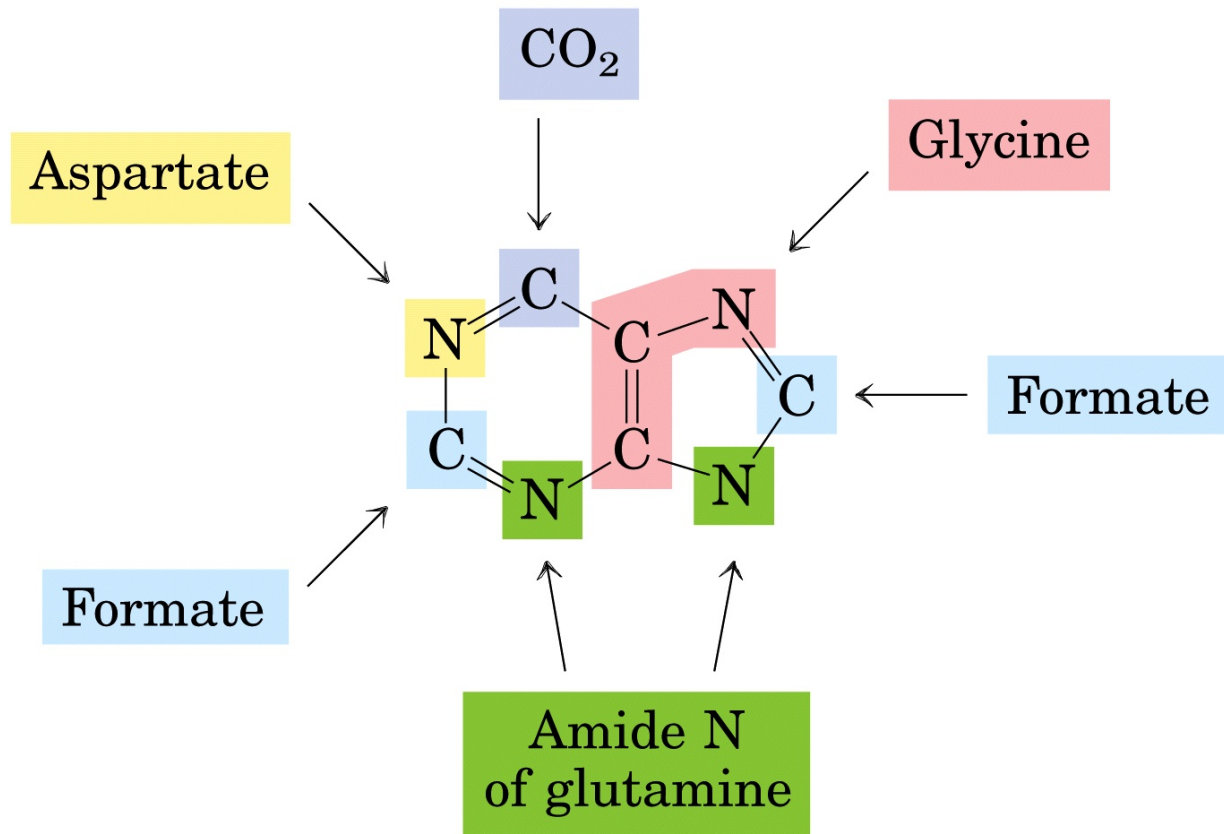


Metabolism of PURINE NUCLEOTIDES

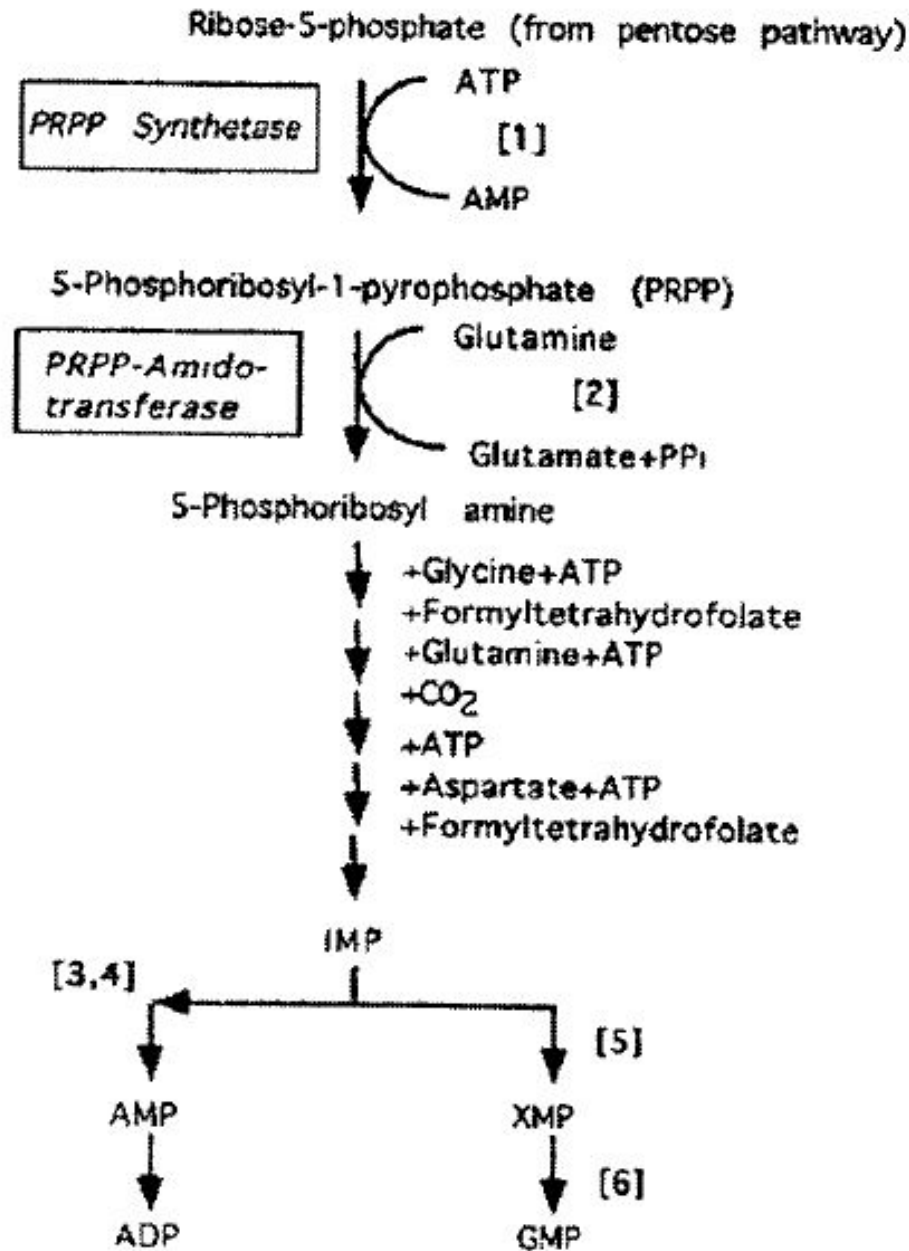
Purine Biosynthesis

De novo: Construct the purine ring afresh

Isotopic labeling experiments defined the precursors.



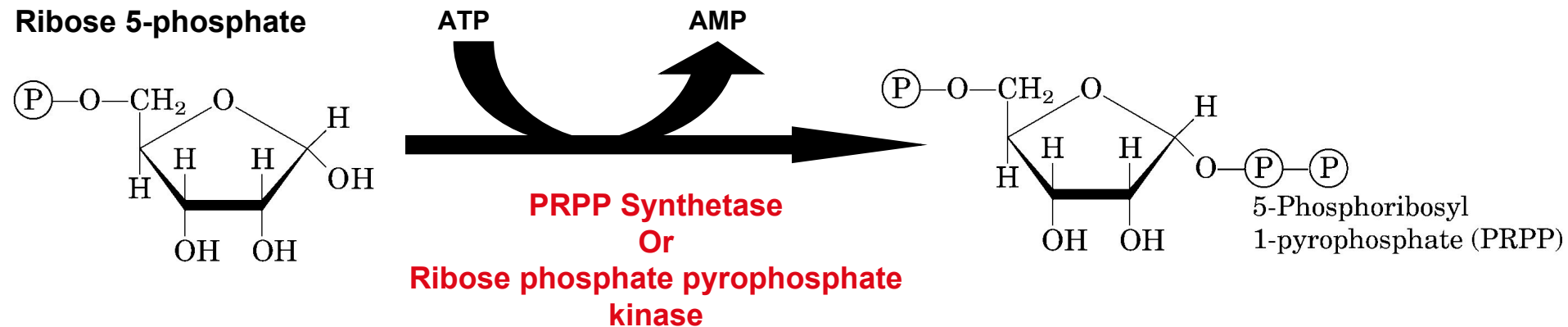
Purine Biosynthesis (Schematic overview)



Purine Biosynthesis

in the beginning there was PRPP

The use of ribose links the pentose phosphate pathway to nucleotide metabolism.



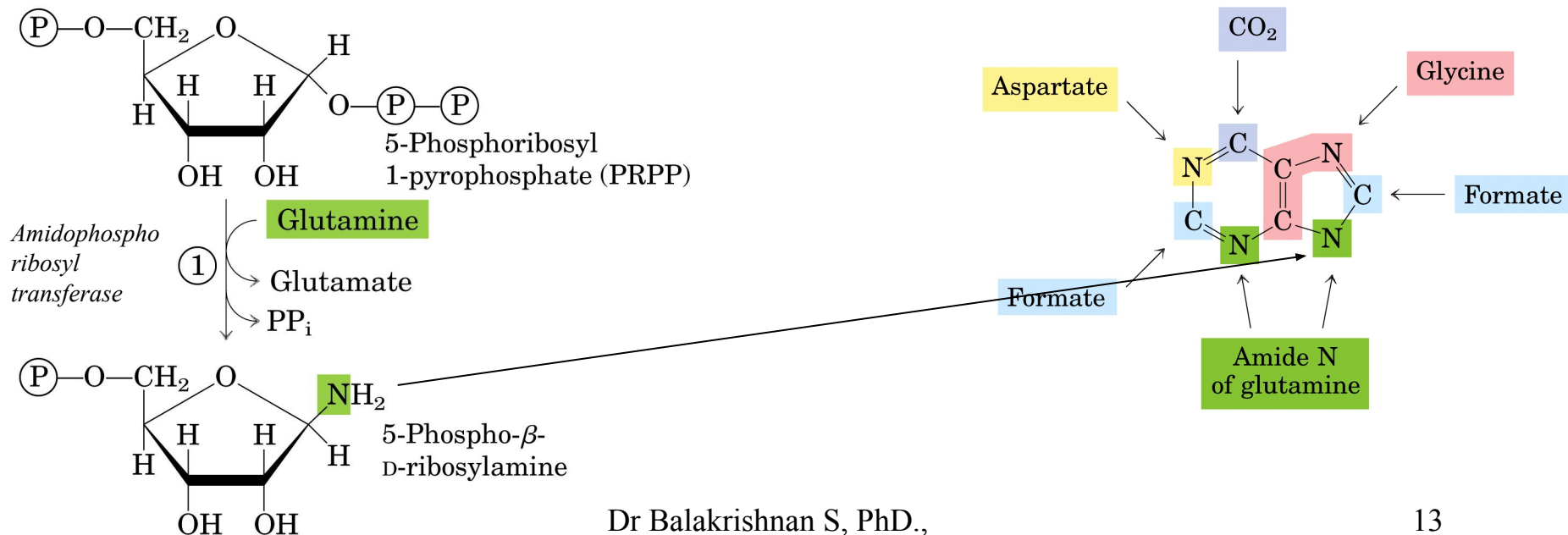
The purine ring is built upon ribose using PRPP

Purine Biosynthesis

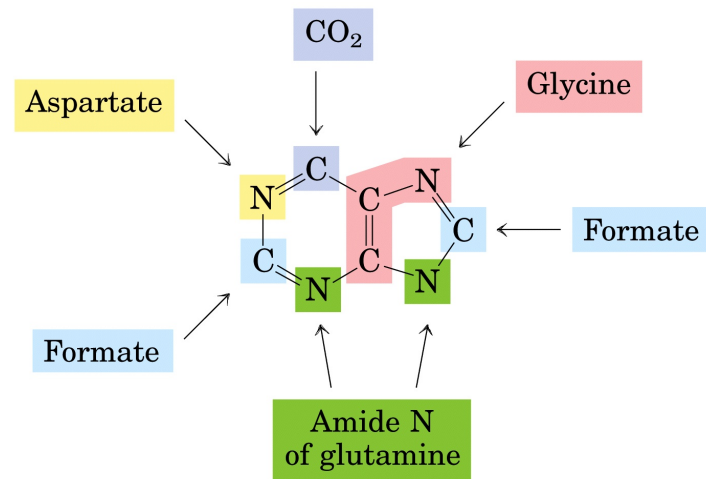
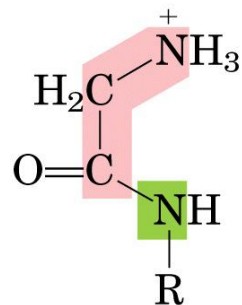
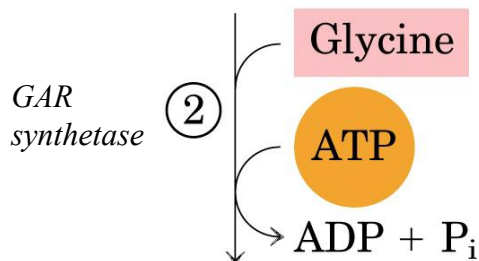
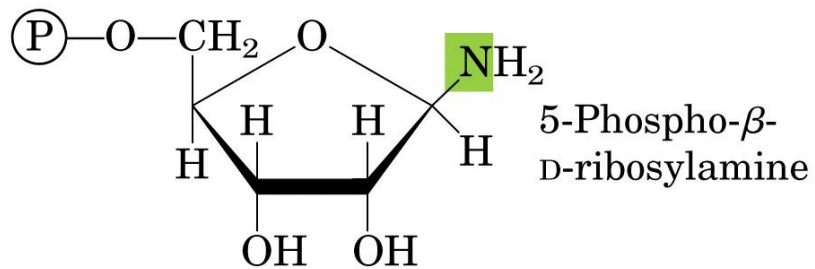
the first committed step

It is always a good idea to know the first committed step of any pathway.

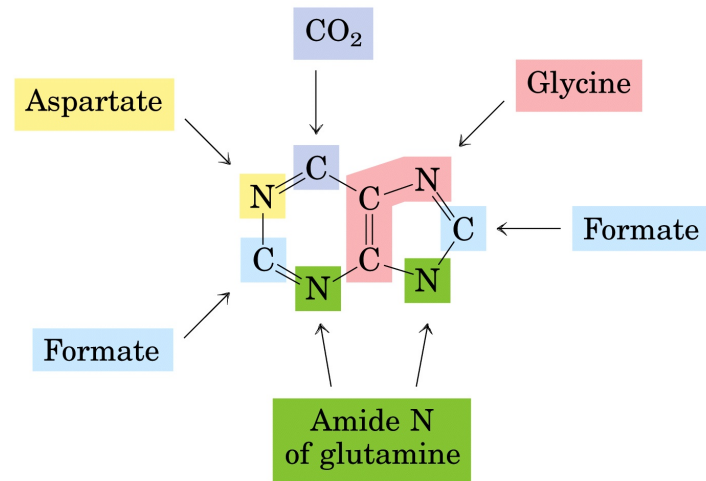
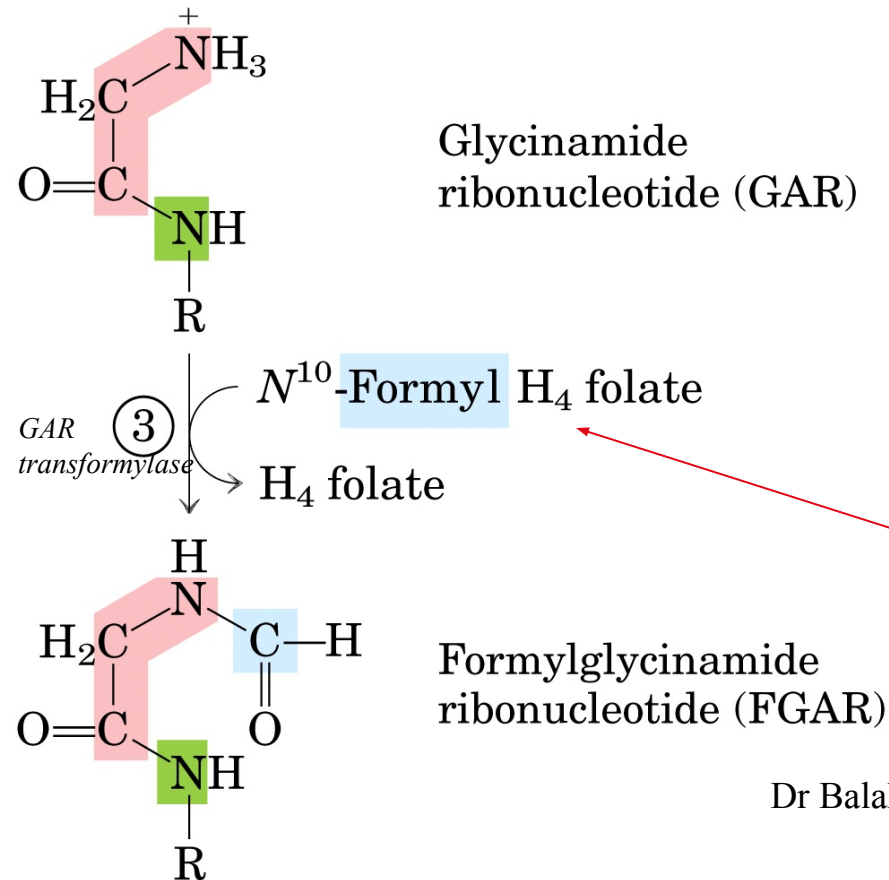
① Glutamine-PRPP amidotransferase



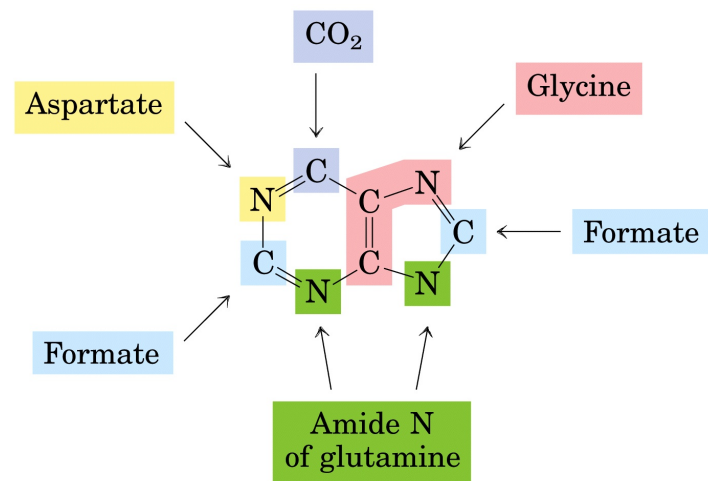
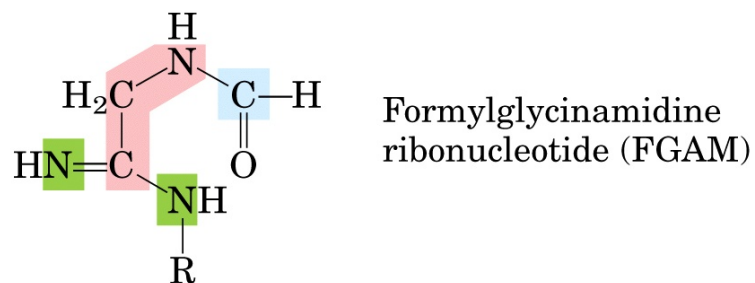
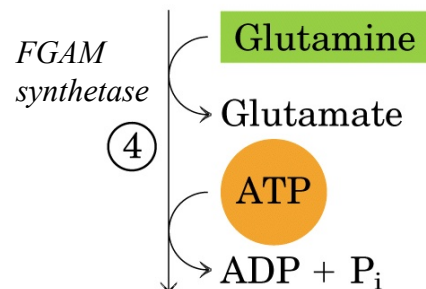
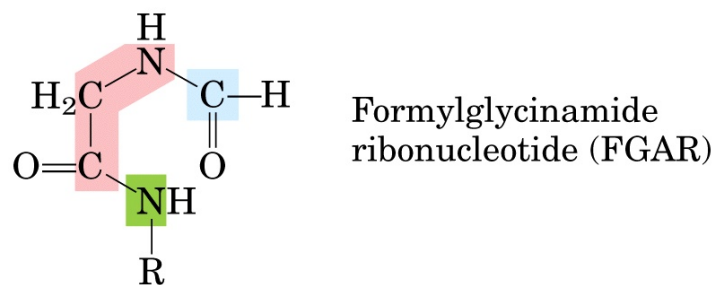
Purine Biosynthesis addition of glycine



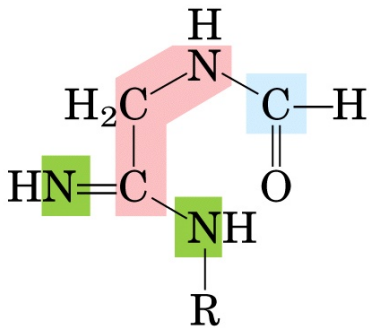
Purine Biosynthesis formylation #1



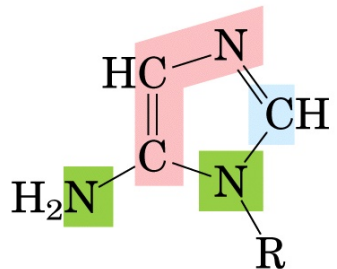
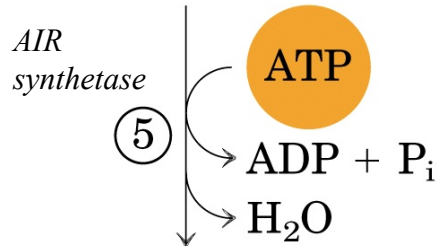
Purine Biosynthesis amidotransferase #2



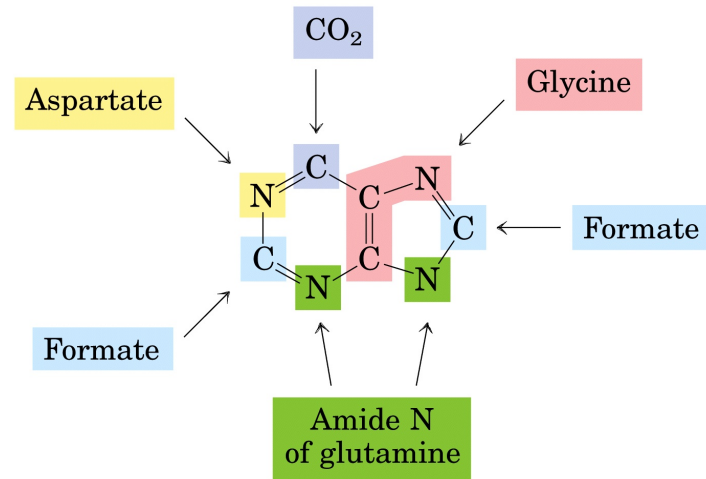
Purine Biosynthesis imidazole ring closure



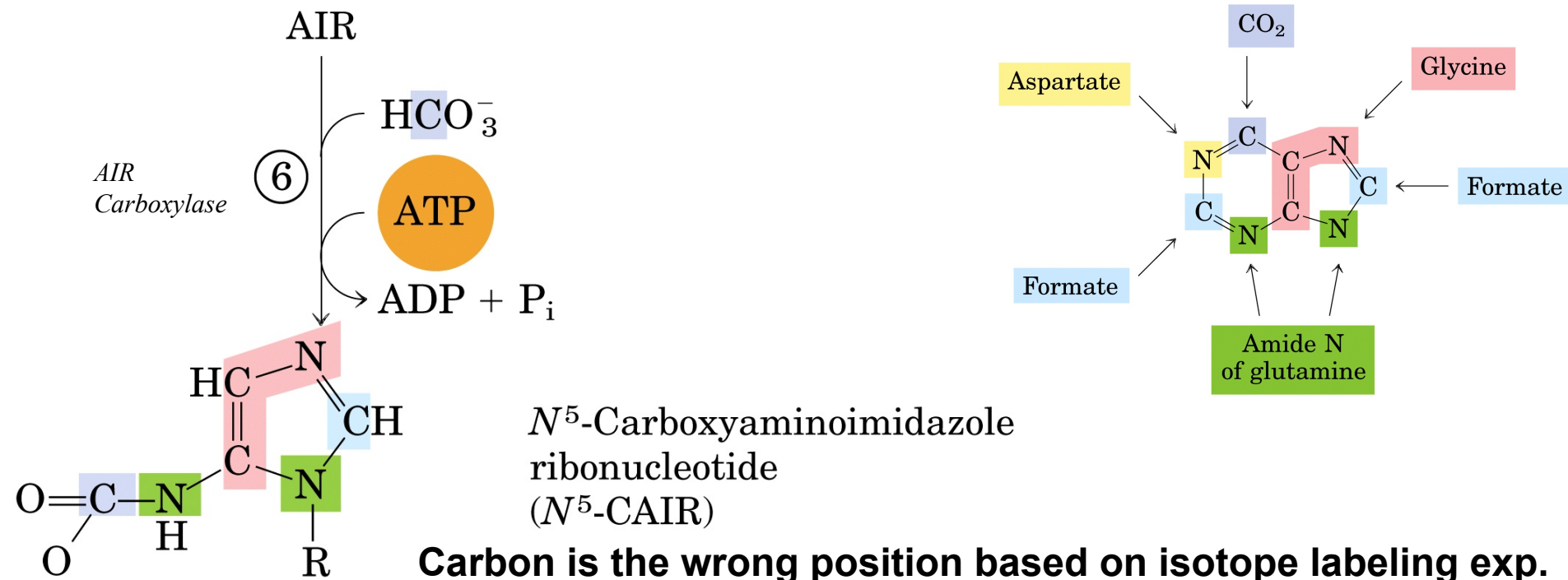
Formylglycinamidine
ribonucleotide (FGAM)



5-Aminoimidazole
ribonucleotide (AIR)

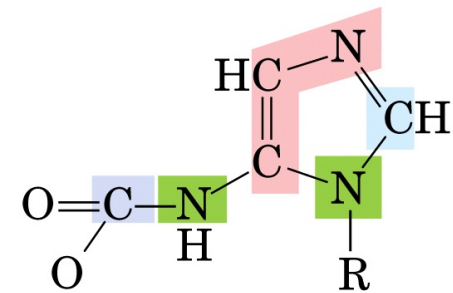


Purine Biosynthesis carboxylation



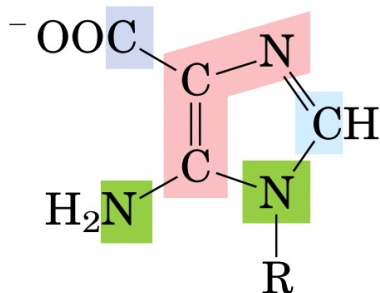
Purine Biosynthesis

move the carbon

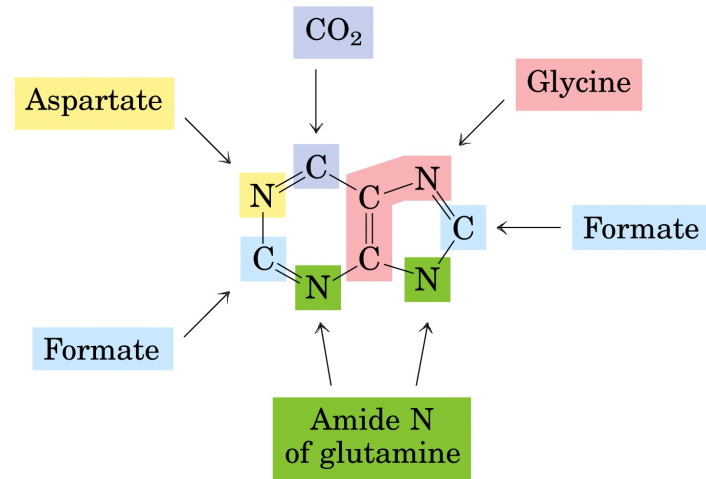


*N*⁵-Carboxyaminoimidazole
ribonucleotide
(*N*⁵-CAIR)

⑦

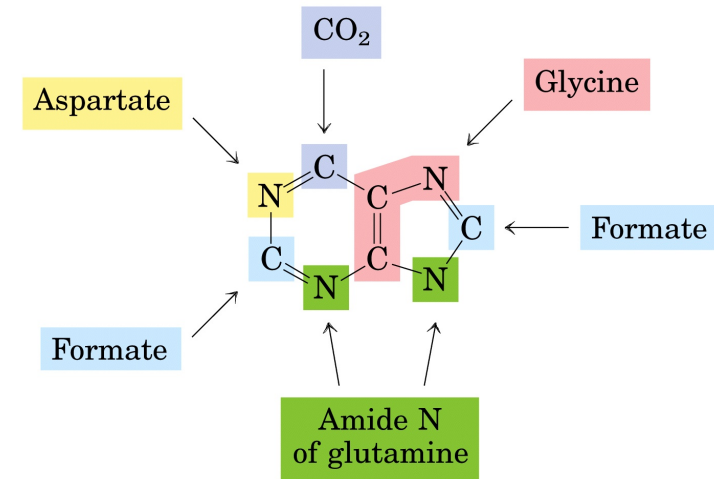
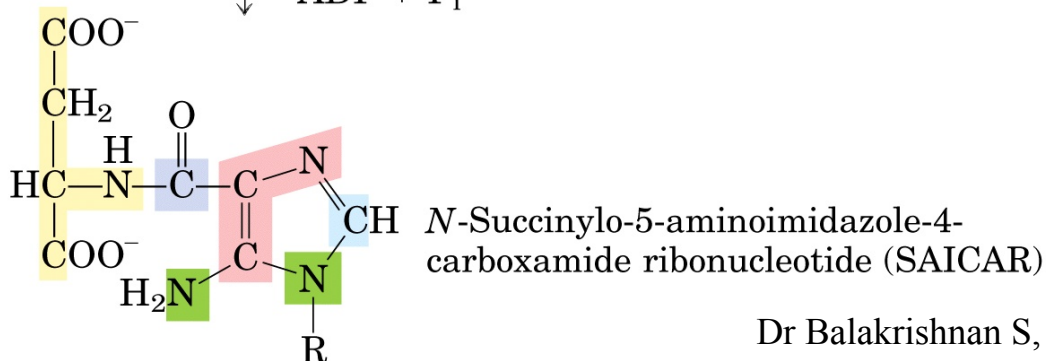
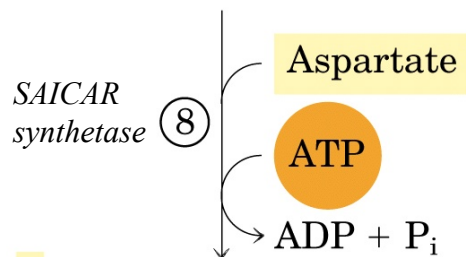
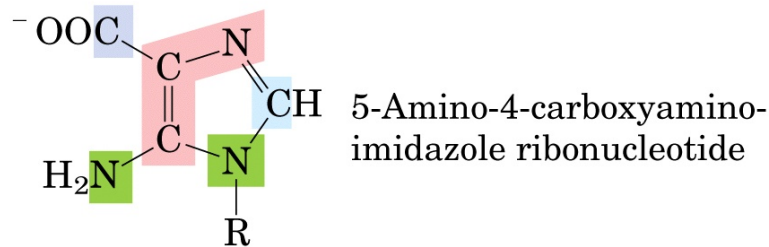


5-Amino-4-carboxyamino-
imidazole ribonucleotide



Now the carbon is placed correctly.

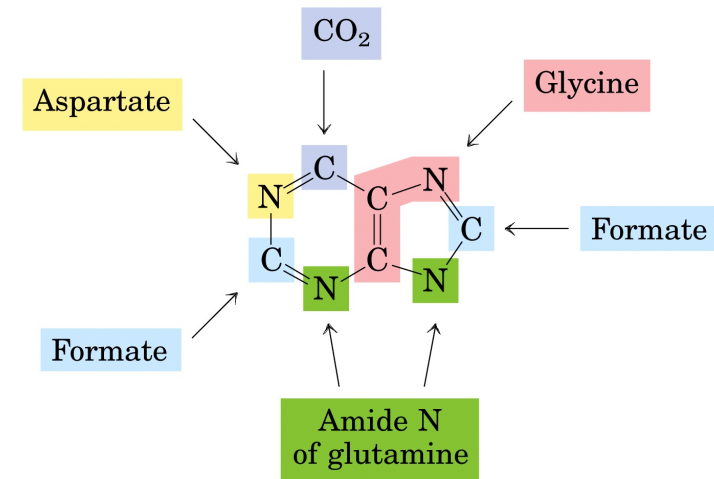
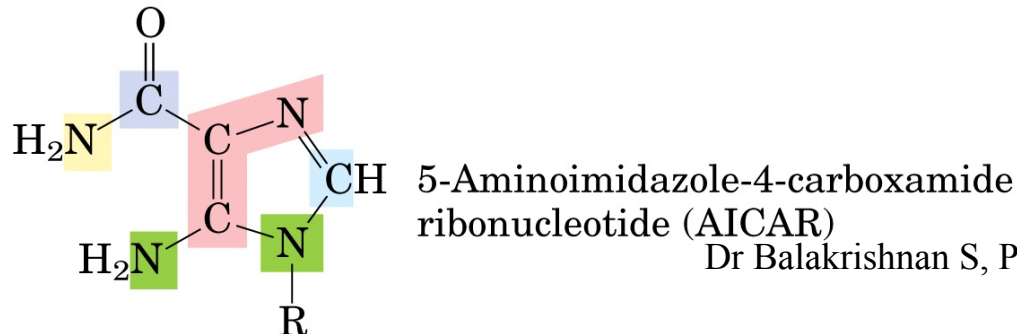
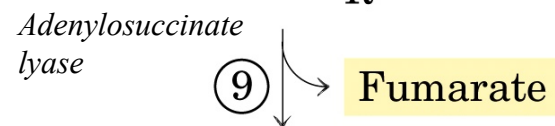
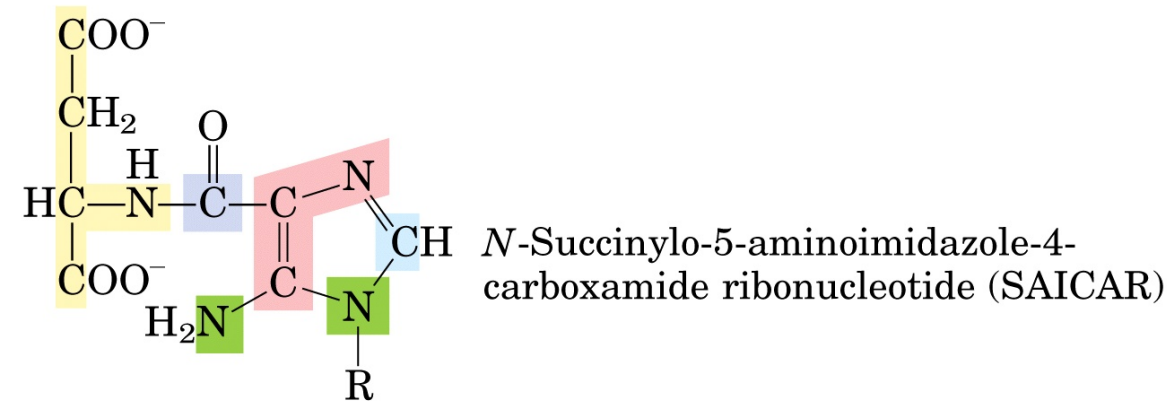
Purine Biosynthesis aspartate addition



But all we need is the nitrogen.

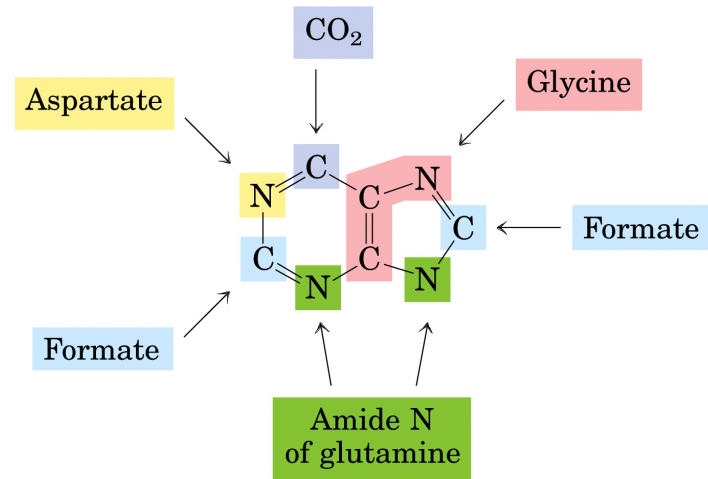
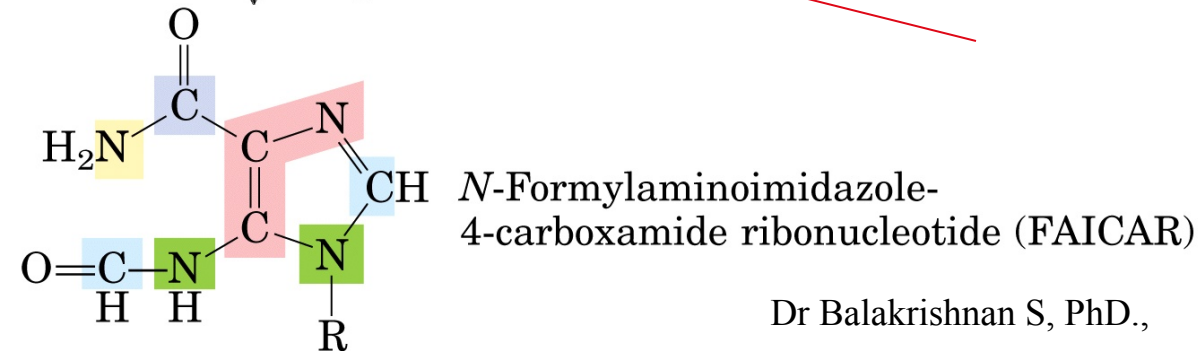
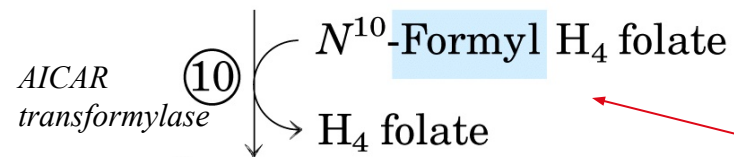
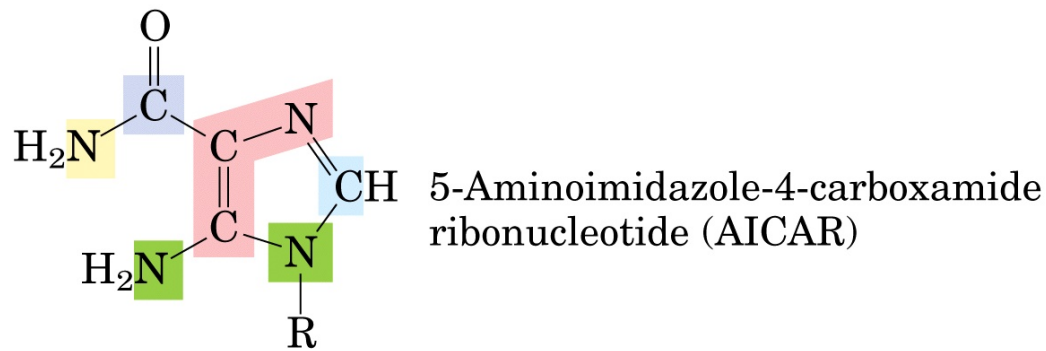
Purine Biosynthesis

fumarate subtraction

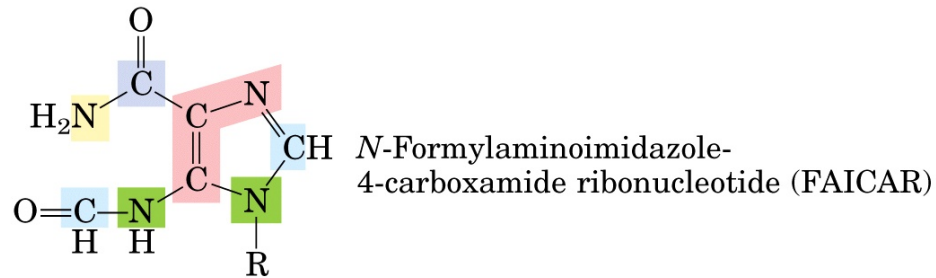


One more carbon.

Purine Biosynthesis formylation #2

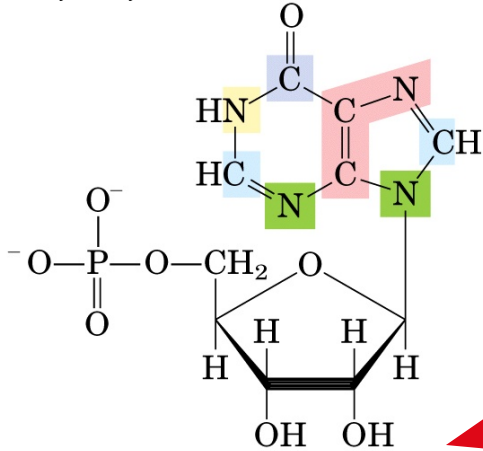


Purine Biosynthesis finally a purine

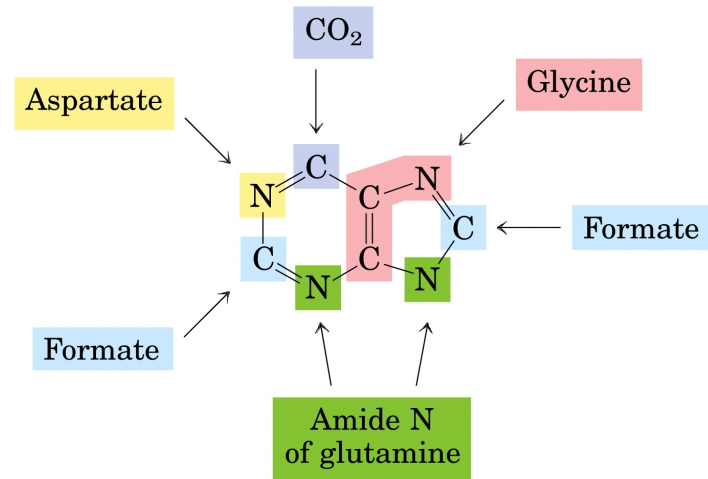


IMP
cyclohydrolase

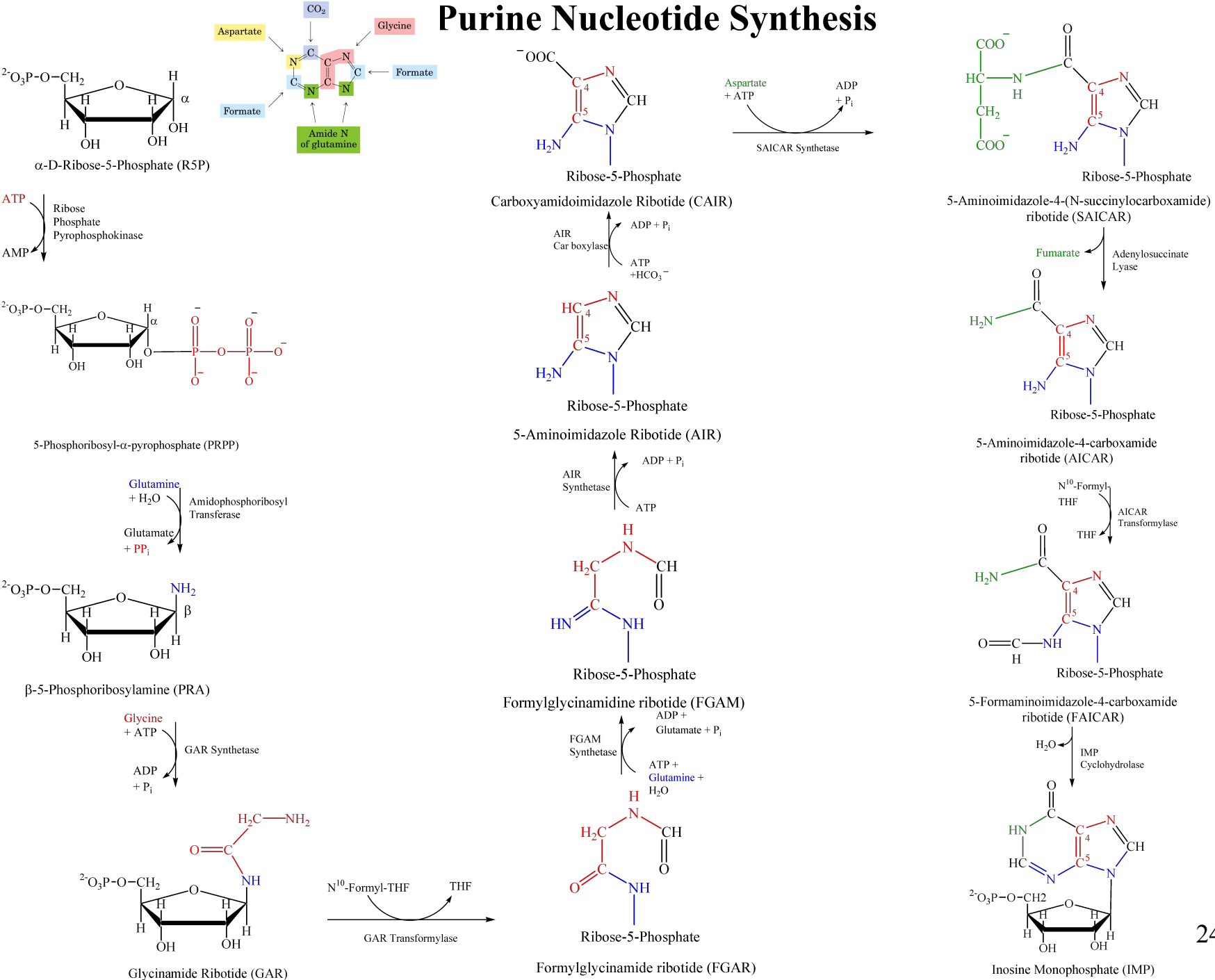
(11) $\rightarrow$ H₂O



Inosinate (IMP)

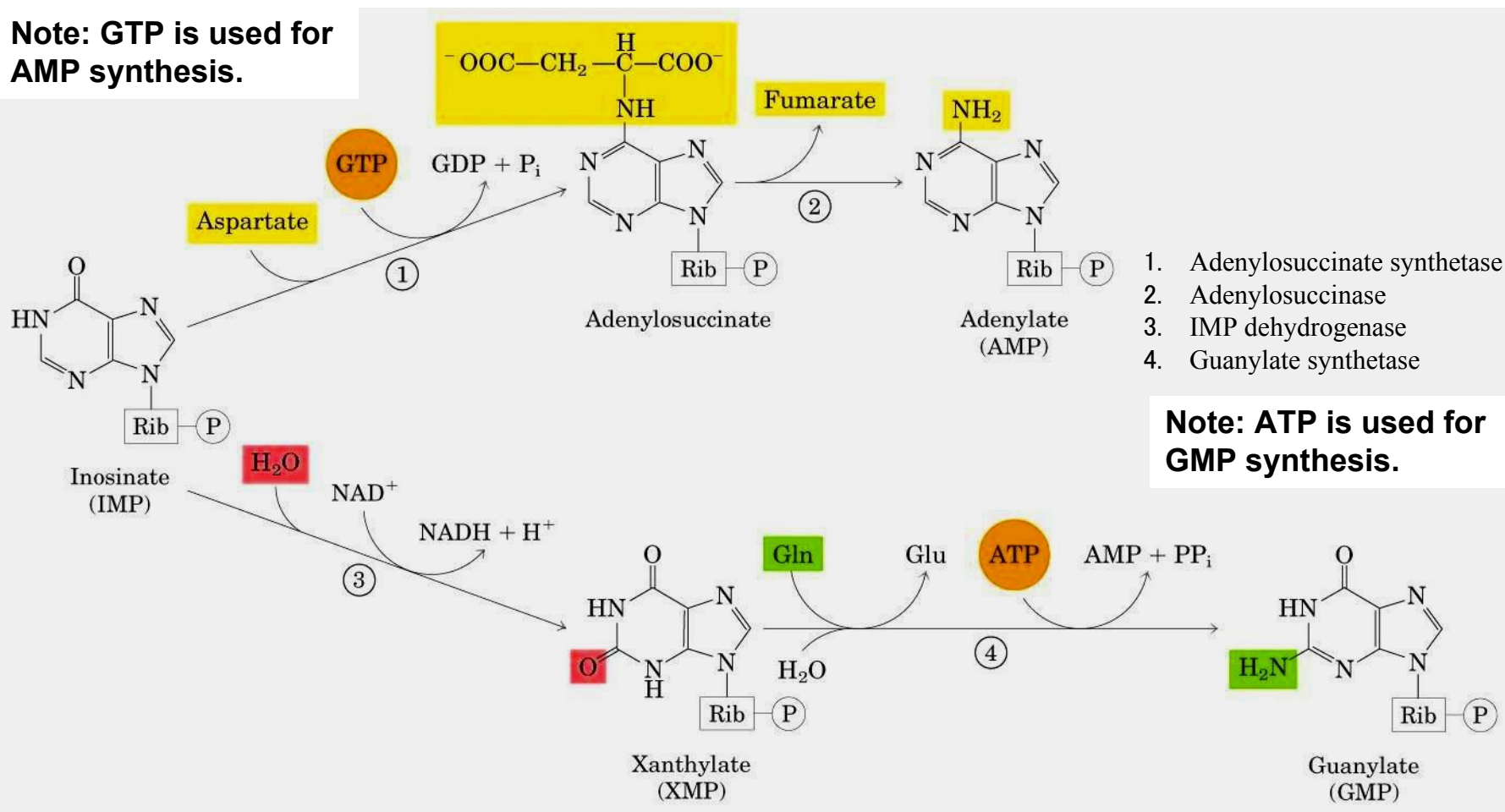


Purine Nucleotide Synthesis



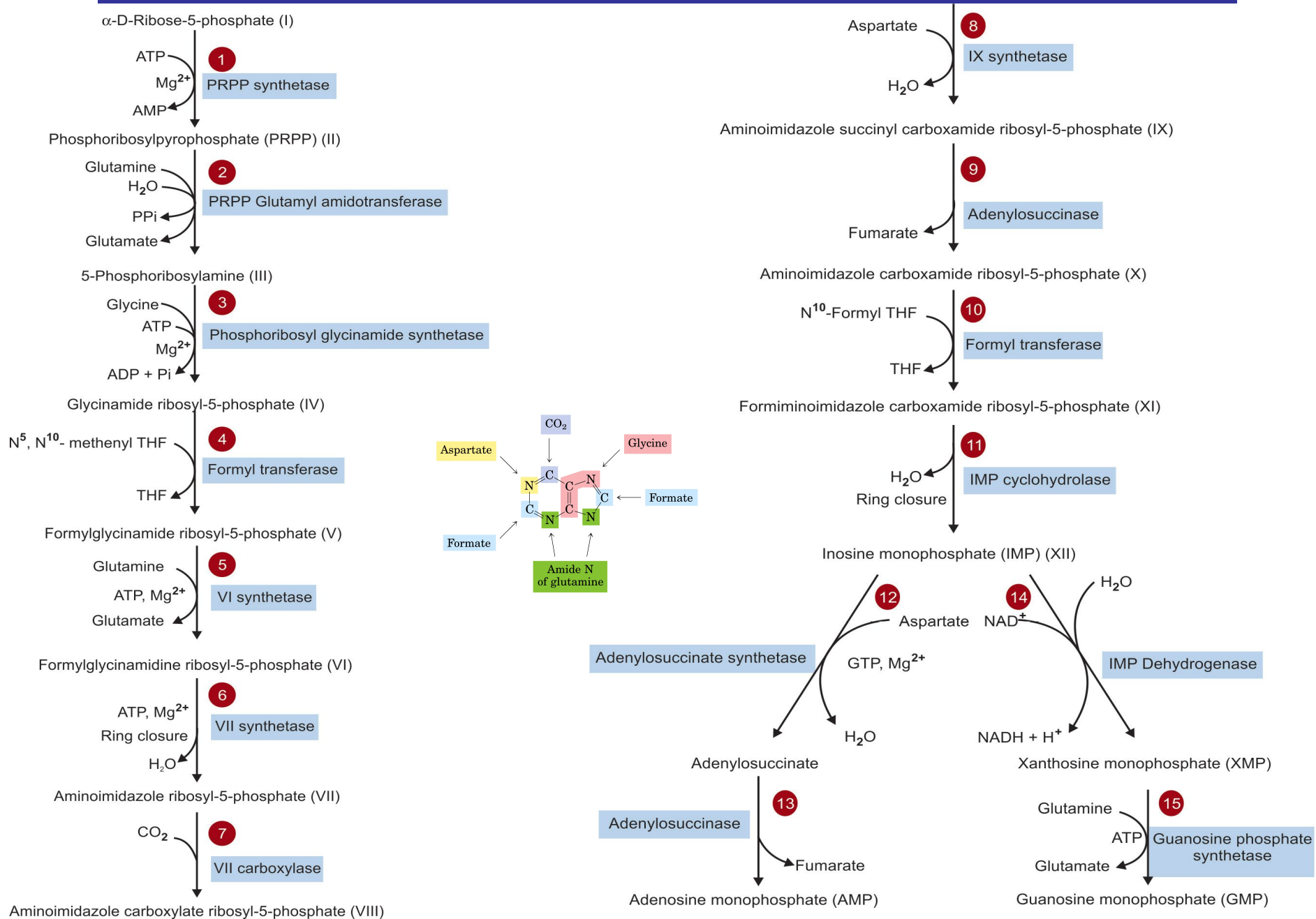
Conversion of IMP to AMP and GMP

Note: GTP is used for AMP synthesis.



Note: ATP is used for GMP synthesis.

Purine Biosynthesis (Schematic Presentation)



Salvage Pathways of Purine Synthesis

- Bases or nucleosides (arising from degradation of preexisting nucleic acid bases) to nucleotides using specific enzymes.

- Adenine phosphoribosyl transferase (APRT)



- Hypoxanthine-Guanine phosphoribosyl transferase (HGPRT)

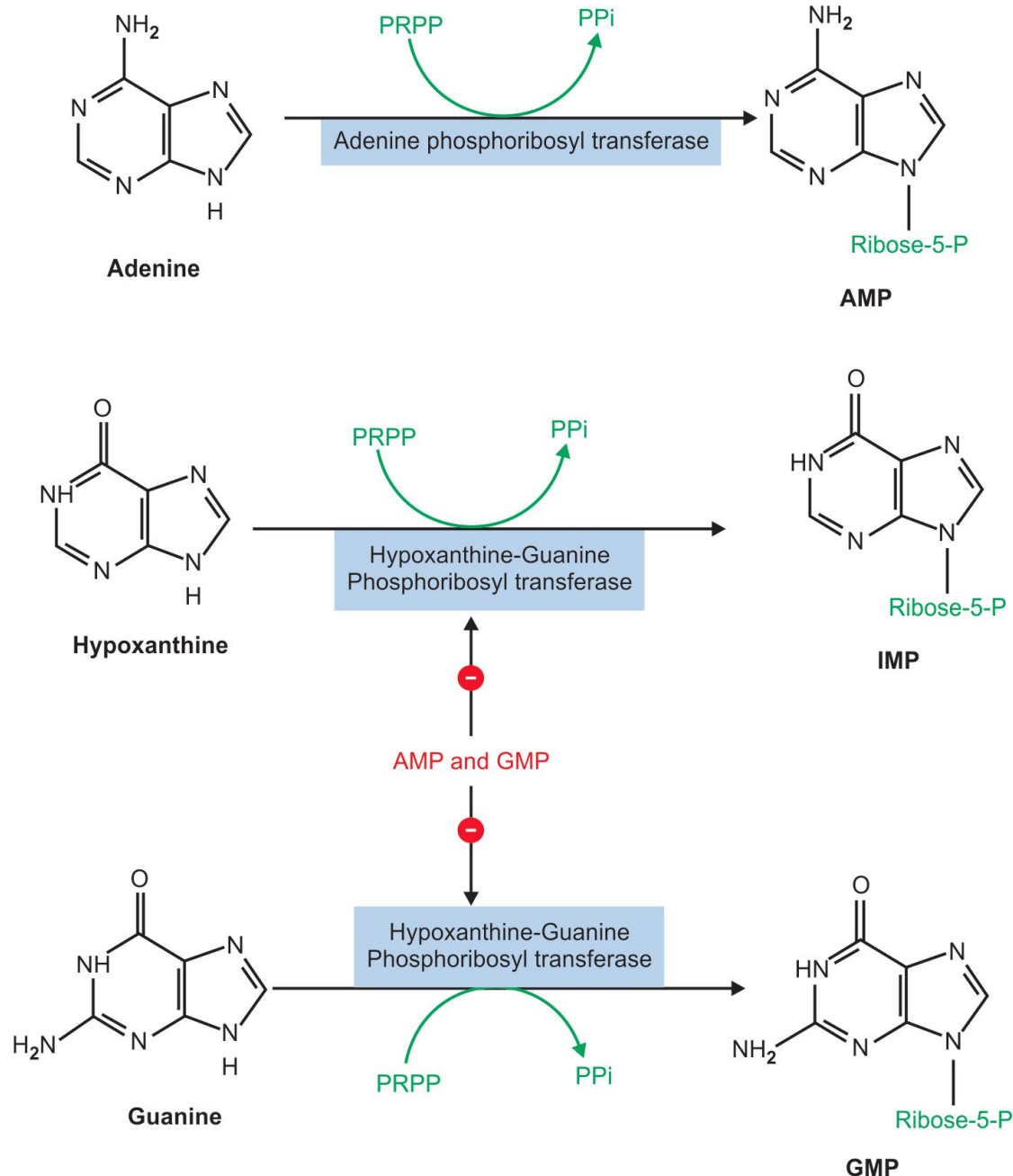


(NOTE: THESE ARE ALL REVERSIBLE REACTIONS)

AMP, IMP, GMP do not need to be resynthesized de novo always!

- Brain, RBCs and polymorphonuclear leucocytes cannot produce purines by de novo synthesis.

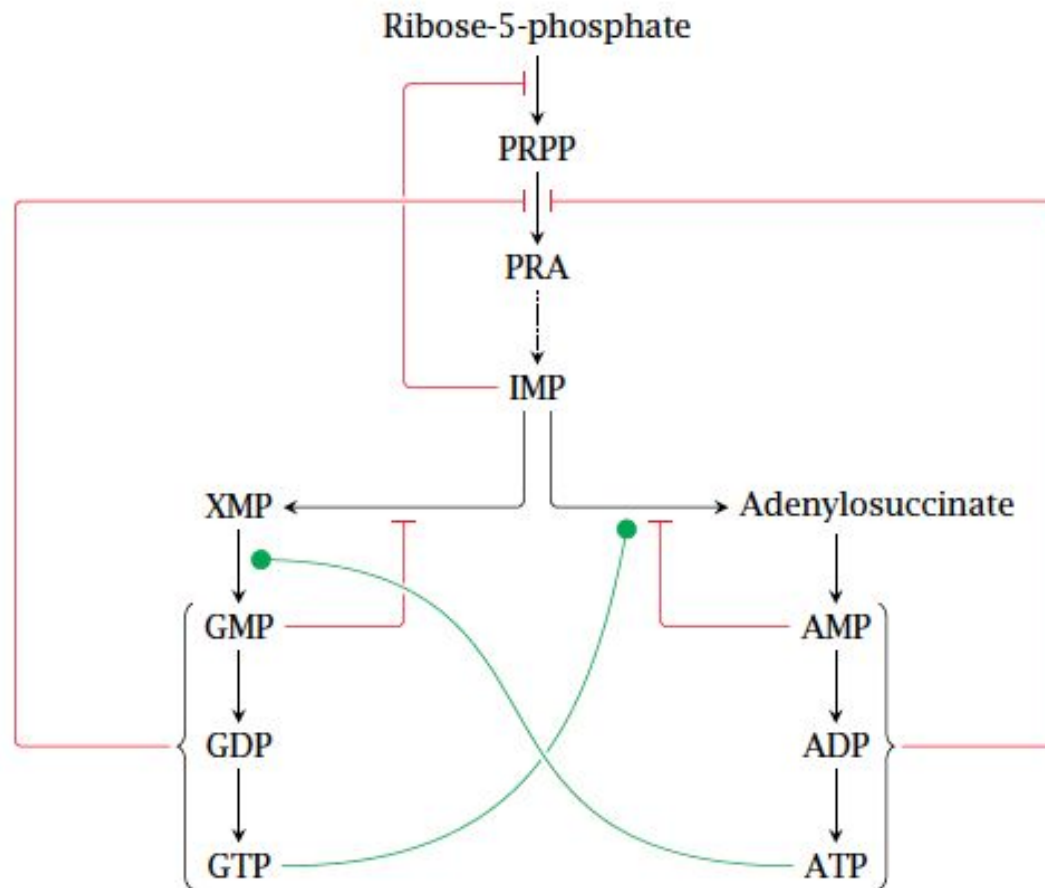
Salvage Pathways of Purine Synthesis



Regulatory Control of Purine Nucleotide Biosynthesis

- PRPP is a biosynthetically “central” molecule (why?)
 - ADP/GDP levels – negative feedback on Ribose Phosphate Pyrophosphokinase
 - Amidophosphoribosyl transferase is activated by PRPP levels
- GTP is involved in AMP synthesis and ATP is involved in GMP synthesis (reciprocal control of production)
- APRT activity has negative feedback at two sites
 - ATP, ADP, AMP bound at one site
 - GTP, GDP AND GMP bound at the other site
- Rate of AMP production increases with increasing concentrations of GTP; rate of GMP production increases with increasing concentrations of ATP

Feedback regulation in purine synthesis



Purine Catabolism and Salvage

- All purine degradation leads to uric acid (but it might not stop there)
- Ingested nucleic acids are degraded to nucleotides by pancreatic nucleases, and intestinal phosphodiesterases in the intestine
- Group-specific nucleotidases and non-specific phosphatases degrade nucleotides into nucleosides
 - Direct absorption of nucleosides
 - Further degradation

Nucleoside + H₂O $\square$ base + ribose (nucleosidase)

Nucleoside + P_i $\square$ base + r-1-phosphate (n. phosphorylase)

NOTE: MOST INGESTED NUCLEIC ACIDS ARE DEGRADED AND EXCRETED.

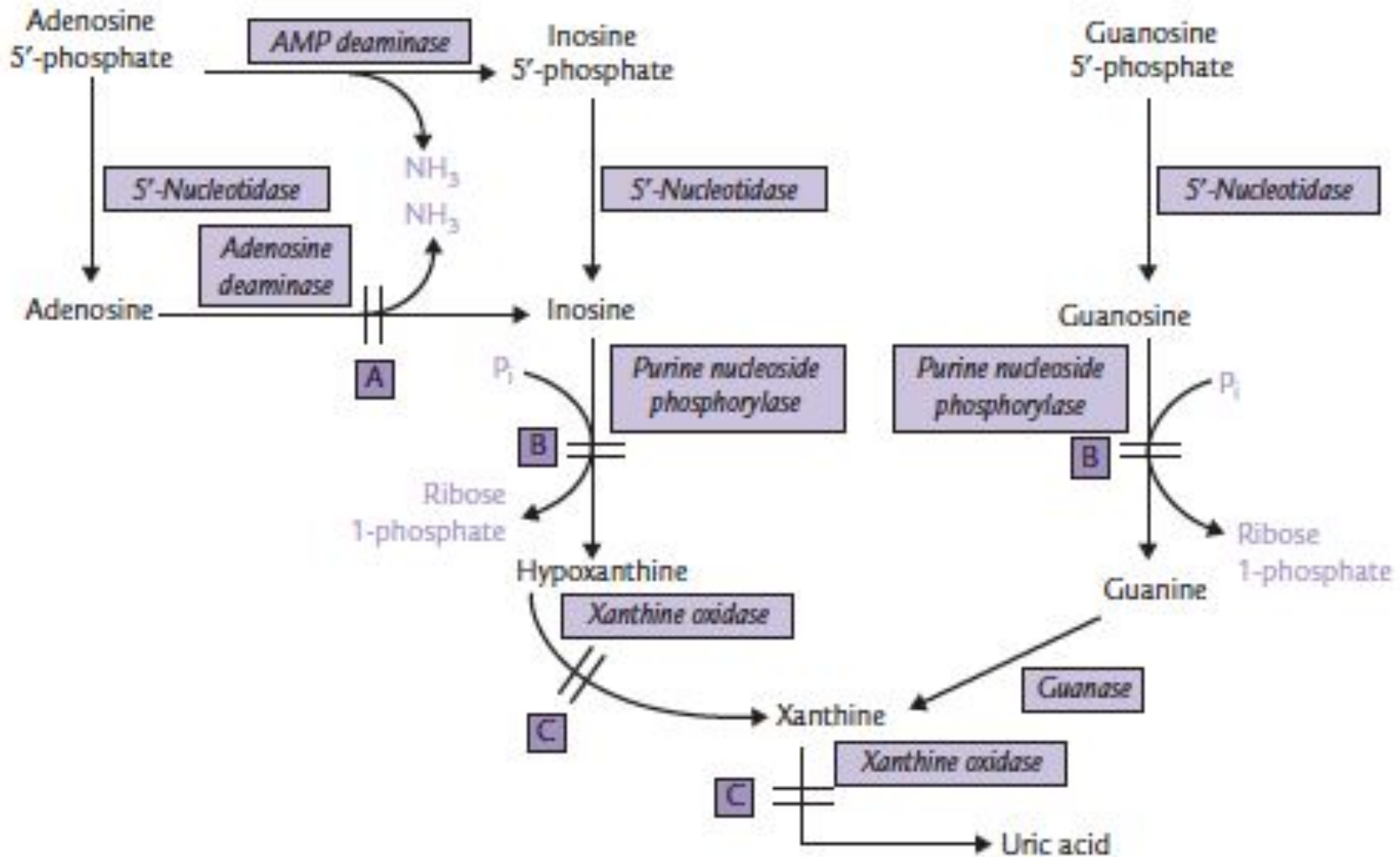
Intracellular Purine Catabolism

- Nucleotides broken into nucleosides by action of 5'-nucleotidase (hydrolysis reactions)
- Purine nucleoside phosphorylase (PNP)
 - Inosine $\square$ Hypoxanthine
 - Xanthosine $\square$ Xanthine
 - Guanosine $\square$ Guanine
 - Ribose-1-phosphate splits off
 - Can be isomerized to ribose-5-phosphate
- Adenosine is deaminated to Inosine (Adenosine Deaminase)

Intracellular Purine Catabolism

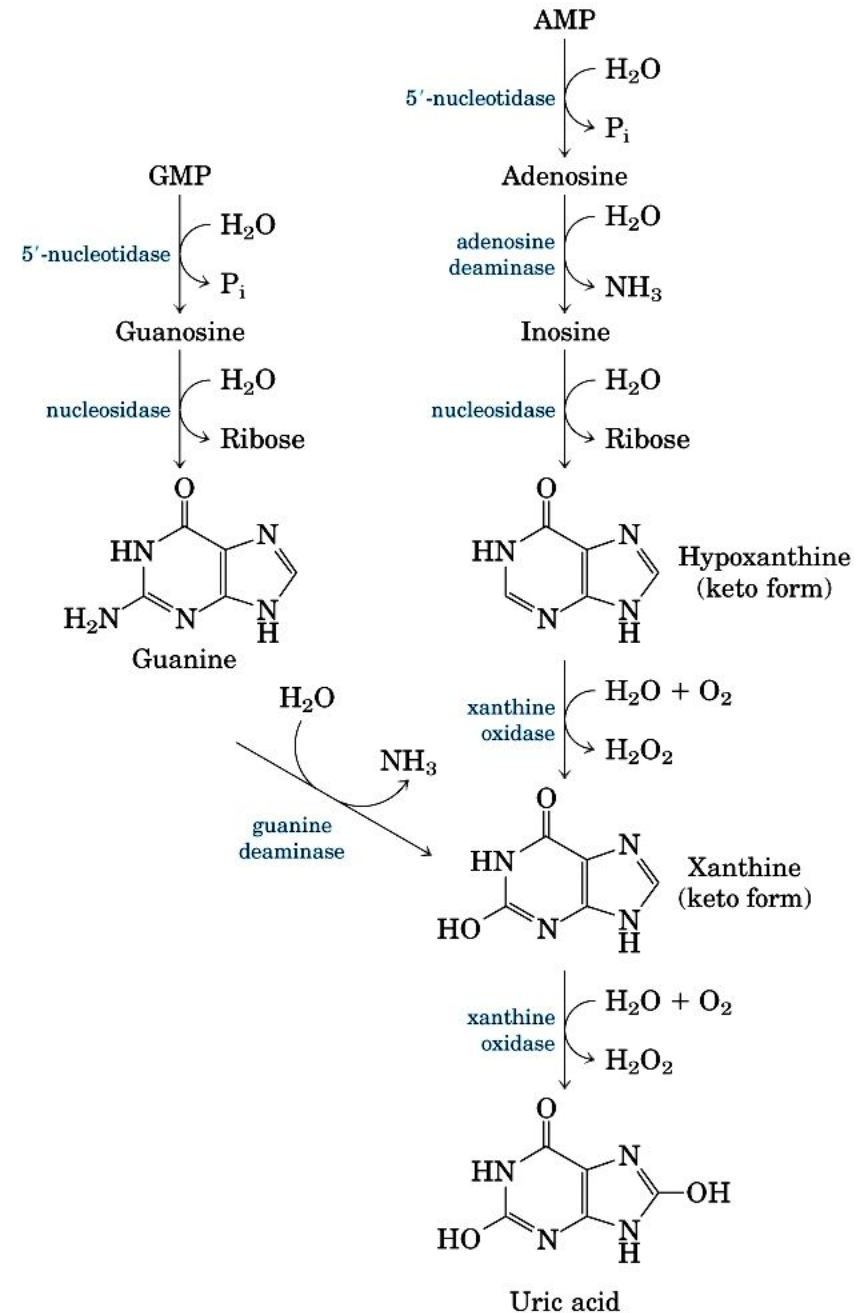
- Xanthine is the point of convergence for the metabolism of the purine bases
- Xanthine $\rightarrow$ Uric acid
 - Xanthine oxidase catalyzes two reactions
- Purine ribonucleotide degradation pathway is same for purine deoxyribonucleotides.

Purine Catabolism



Purine Catabolism/ Formation of uric acid

1. Remove phosphate.
-5'-nucleotidase
- 1a. Deaminate adenosine
-Adenosine deaminase
2. Remove the ribose.
-nucleosidase
3. Generate xanthine.
-oxidize hypoxanthine
-deaminate guanine
4. Make uric acid.
-xanthine oxidase

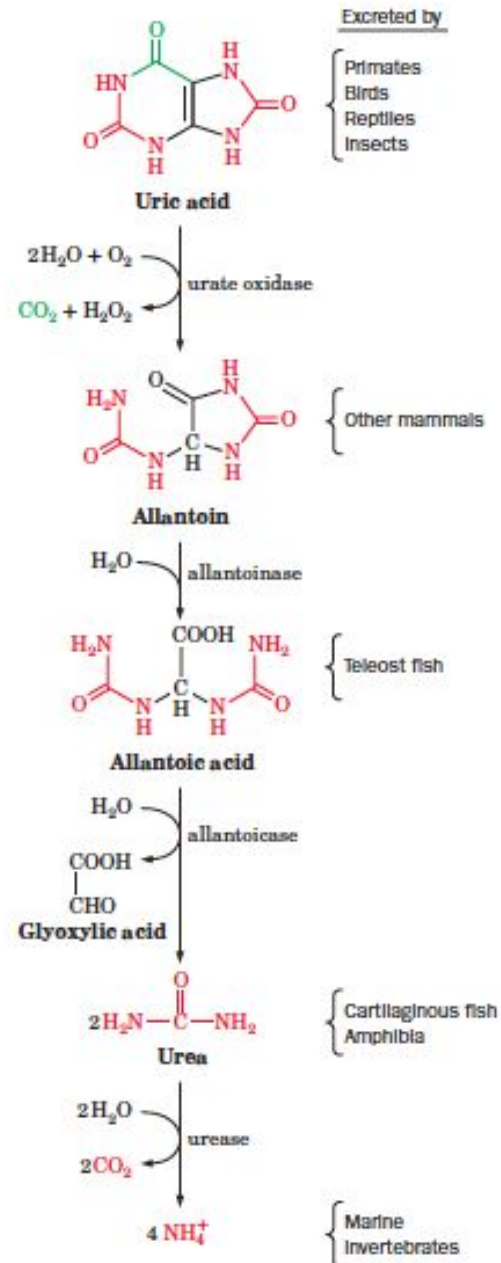


Uric Acid Excretion

- Humans – excreted into urine as insoluble crystals
- Birds, terrestrial reptiles, some insects – excrete insoluble crystals in paste form
 - Excess amino N converted to uric acid
 - (conserves water)
- Others – further modification :

Uric Acid ☐ Allantoin ☐ Allantoic Acid ☐ Urea ☐ Ammonia

Degradation of uric acid to ammonia



Hyperuricemia

- An elevated serum urate (uric acid) concentration - hyperuricemia.
 - The catabolism of the purines (adenine and guanine) produces uric acid.
 - At physiological pH, uric acid is mostly ionized and present in plasma as sodium urate.
- Uric acid and urate are relatively insoluble molecules - Precipitate out of aqueous solutions such as urine or synovial fluid.
 - Found primarily in the joints of the extremities and in the renal interstitium.
 - The consequence of this is the condition, gout.
 - Extreme swelling and tenderness in the joints of the extremities due to the presence of urate crystals.
- The average normal blood serum level of uric acid is 4 to 7 mg per 100 ml.

Hyperuricemia (Cont'd)

- Diagnosis: $\square$ uric acid - detectable in the blood and urine.
- Definitive diagnosis: Presence of urate crystals in synovial fluid - collected from affected joints.
- The urate crystal formation in the joints of the extremities, such as the big toe, is associated with the decreased temperature of the extremities.
 - Low temp aids in urate crystal formation when levels exceed solubility.

Gout

- Impaired excretion or overproduction of uric acid
- Uric acid crystals precipitate into joints (Gouty Arthritis), kidneys, ureters (stones)
- Uric acid is deposited in cooler areas of the body to cause tophi.
 - The presence of urate crystals is associated with extreme swelling and tenderness in the joints of the extremities
- The total urate pool (normal 1200 mg) is increased to 3000 mg in gout patients without tophi. It may be as high as 30,000 mg in patients with tophi.





Classification of gout

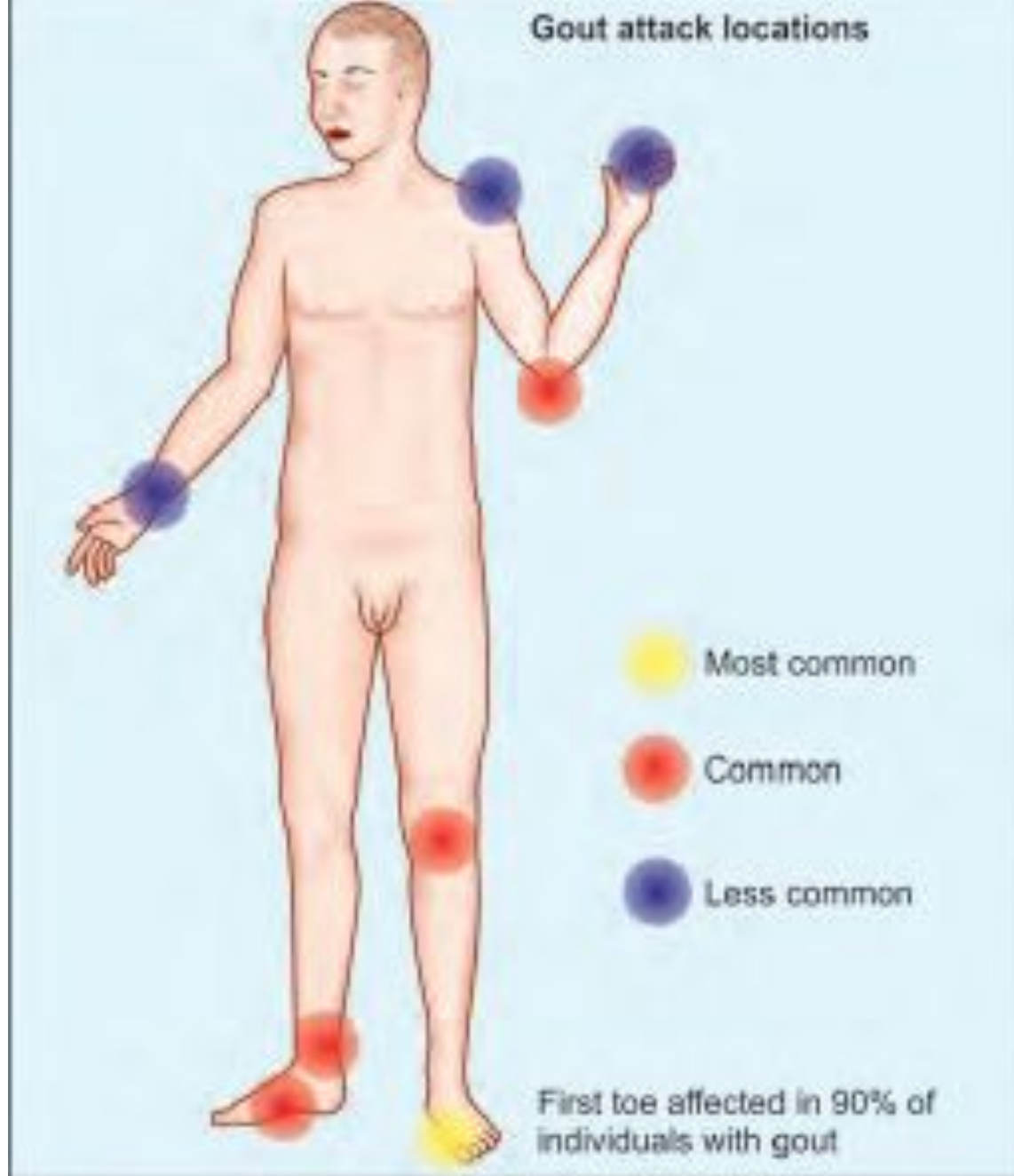
- Classified into two broad types:
 1. Primary gout
 2. Secondary gout

Primary gout

- An inborn error of metabolism due to overproduction of uric acid.
- Increased level of uric acid is associated with increased synthesis of purine nucleotides;
 - Caused by following defective enzymes:
 - PRPP synthetase
 - PRPP glutamyl amidotransferase
 - HGPRTase
 - Glucose-6-phosphatase

Secondary gout

- Results from a variety of diseases that cause an elevated destruction of cells or decreased elimination of uric acid;
 - Elevated destruction of cells is accompanied by increased degradation of nucleic acids to uric acid, which occurs in;
 - Cancers (leukemia, polycythemia),
 - Psoriasis and
 - Hyper-catabolic states (starvation, trauma, etc.).
 - Decreased elimination of uric acid occurs in chronic renal disease due to reduced glomerular filtrate rate.

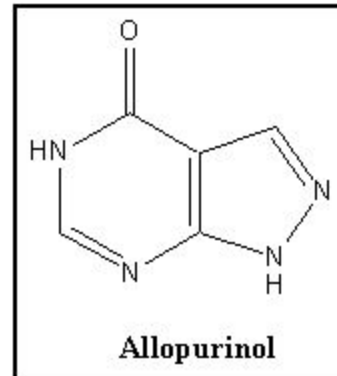
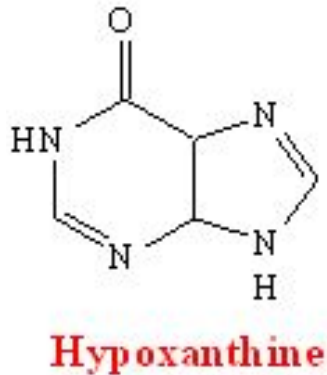


Clinical Findings of Gout

- The joints are extremely painful.
- The typical gouty arthritis affects the first metatarsophalangeal joint (big toe), but other joints may also be affected.
- Synovial fluid will show birefringent urate crystals.
- In chronic cases, uric acid may get deposited around joints causing swelling (tophi) composed of sodium urate.
- In chronic gout, the deposition of urate crystals in renal medulla occurs which progresses to urolithiasis and renal damage.
- Xanthine oxidase inhibitors inhibit production of uric acid, and treat gout
 - Allopurinol treatment – hypoxanthine analog that binds to Xanthine Oxidase to decrease uric acid production

ALLOPURINOL: A XANTHINE OXIDASE INHIBITOR

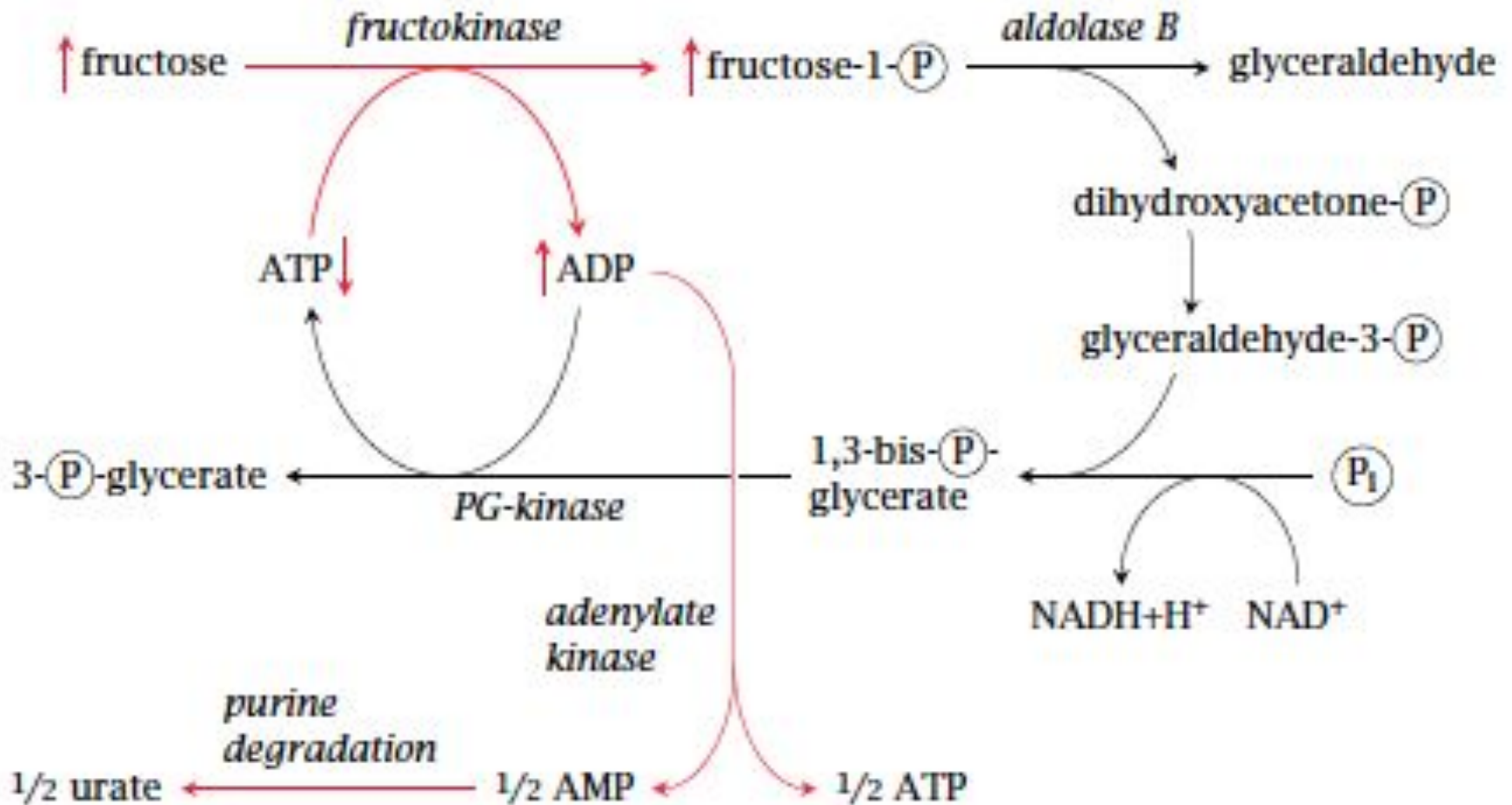
- Allopurinol, a substrate analog is converted to an inhibitor (suicide-inhibitor), is used to treat chronic gout.



Diets and drugs that may promote gout

- Too much food too rich in purines
- Excessive fructose or sucrose
- Alcoholism—Purine contents and lactate formation
- Lead poisoning - lead impairs uric acid excretion
- Drugs that interfere with uric acid secretion: pyrazinamide, salicylic acid
- Drugs that contain purines: Adefovir dipivoxil (a nucleotide analog), Acyclovir (a guanosine analog), dideoxyadenosine

Gout: the fructose connection



Pseudogout

- Deposition of calcium pyrophosphate crystals in joints leads to pseudogout.
- Clinical manifestations:
 - Similar to those in gout
 - But serum uric acid is not elevated.

Lesch-Nyhan Syndrome

- An X-linked inherited disorder.
- A complete deficiency of HGPRT
 - Causes increased level of Hypoxanthine and Guanine (□ ↑ degradation to uric acid)
- Also, PRPP accumulates
 - Stimulates production of purine nucleotides (and thereby increases their degradation)
- Causes gout-like symptoms, but also neurological symptoms (psychomotor delay) □ spasticity, aggressiveness, self-mutilation (Compulsive self-injurious behaviour)
 - Due to extremely low GTP, which is required for activation of GABA
- First neuropsychiatric abnormality that was attributed to a single enzyme

Kelly-Seegmiller Syndrome

- An X-linked inherited disorder.
- A partial deficiency of HGPRT
- Associated with gout, but not neurological involvement
 - Also called ‘HGPRT related gout’.

Hypouricaemia

- Decreased concentration of serum uric acid (below 2 mg/dL).
- Mostly due to enzymatic defects:
 - Adenosine deaminase (or)
 - Xanthine oxidase
- Less commonly due to;
 - Some renal defect, e.g. Fanconi syndrome (generalized proximal tubules dysfunction; inadequate reabsorption)
- Familial Renal Hypouricaemia:
 - Failure to reabsorb the filtered uric acid by renal tubules due to a genetic defect in urate transporter (URAT1).

ADENOSINE DEAMINASE (ADA) DEFICIENCY

- ADA catalyzes: Deamination of Adenosine □ Inosine, and Deoxyadenosine □ deoxyinosine.
- ADA deficiency results;
 - Accumulation of adenosine and deoxyadenosine
 - The accumulated deoxyadenosine is converted to dAMP, dADP and dATP.
 - The last one (dATP) is a powerful inhibitor of ribonucleotide reductase .
 - Increased amount of deoxyadenosine is toxic to lymphocytes and suppresses immune function □ “Severe Combined Immunodeficiency (SCID)”
 - Selectively kills lymphocytes: Both B- and T-cells

Xanthinuria

- Rare disorder
- Deficiency of xanthine oxidase.
 - Defect in the production of urate,
 - Causes the purine metabolism to stop at xanthine and hypoxanthine.
 - Urate levels in blood and urine are exceedingly low
 - In most cases this condition goes unnoticed.
 - Diagnosis is usually made after the incidental finding of low serum urate.

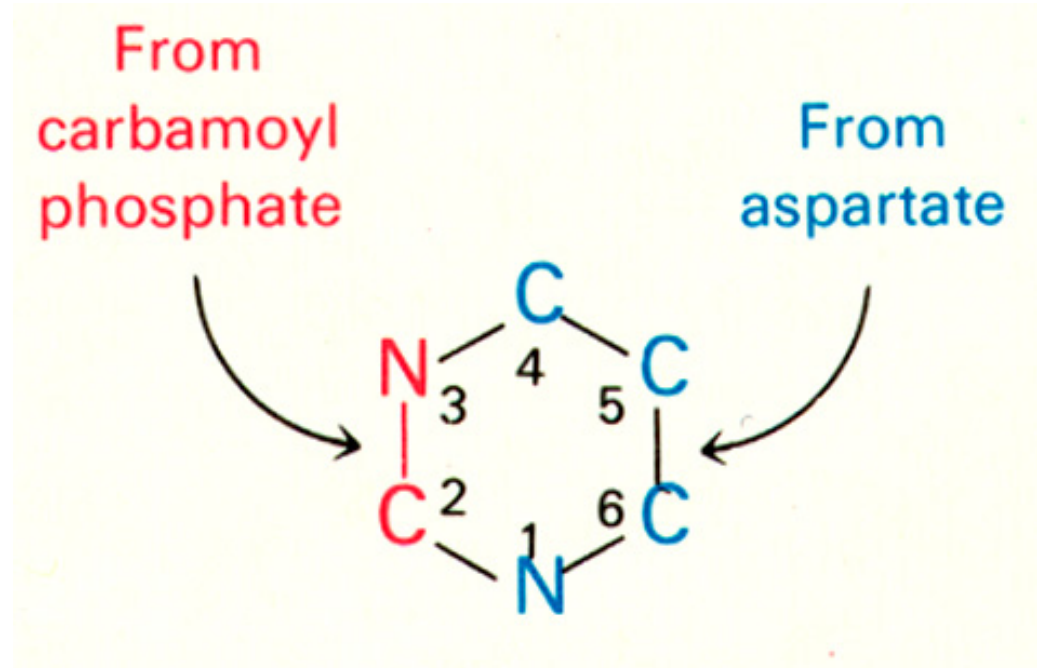
Purine Autism

- 25% of autistic patients may overproduce purines
- To diagnose, must test urine over 24 hours
 - Biochemical findings from this test disappear in adolescence
 - Must obtain urine specimen in infancy, but it's difficult to do!
 - Pink urine due to uric acid crystals may be seen in diapers

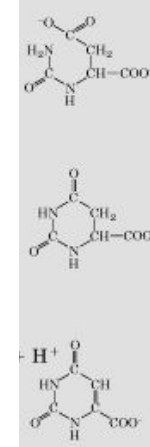
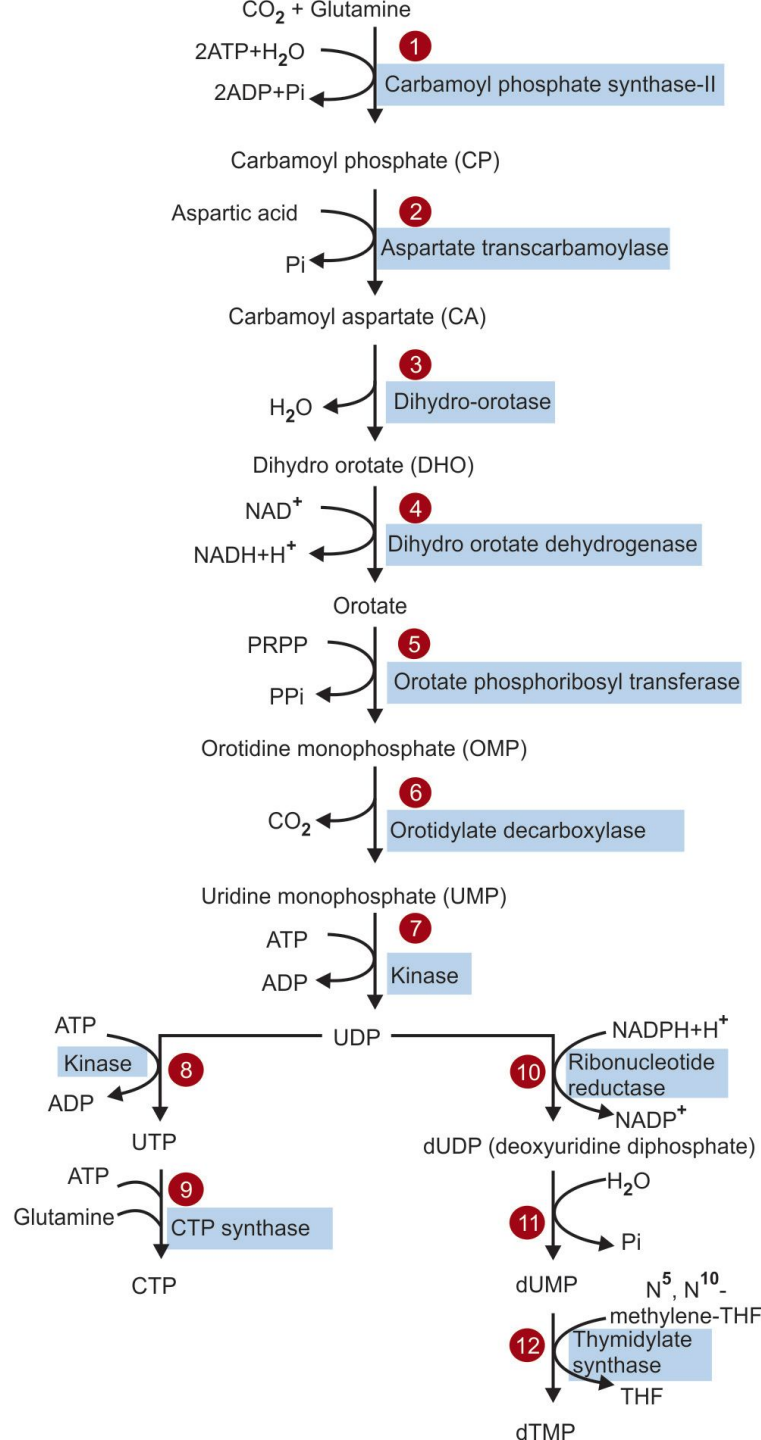
Metabolism of PYRIMIDINE NUCLEOTIDES

Pyrimidine Ribonucleotide Synthesis: de novo

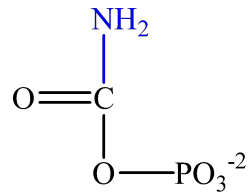
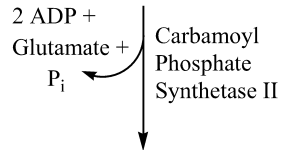
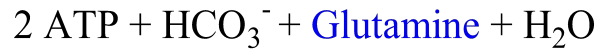
- Pyrimidine ring synthesis completed first; then attached to ribose-5-phosphate
- Uridine Monophosphate (UMP) is synthesized first
- CTP is synthesized from UMP



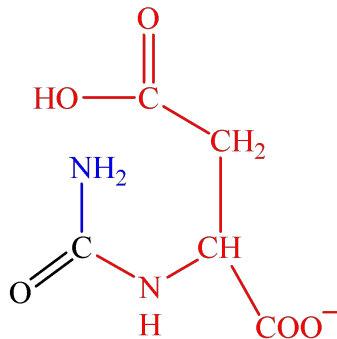
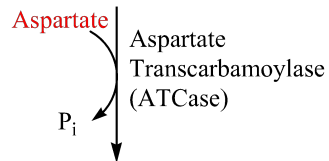
De novo synthesis of Pyrimidine Nucleotides (Schematic presentation)



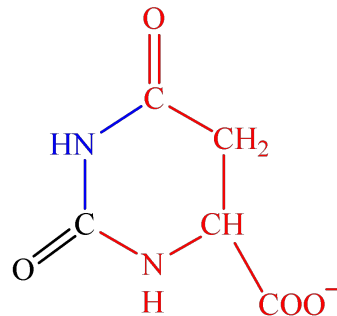
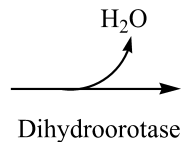
Pyrimidine Synthesis



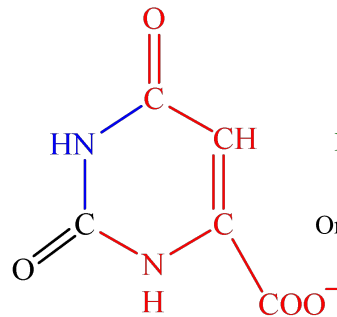
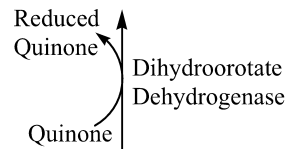
Carbamoyl Phosphate



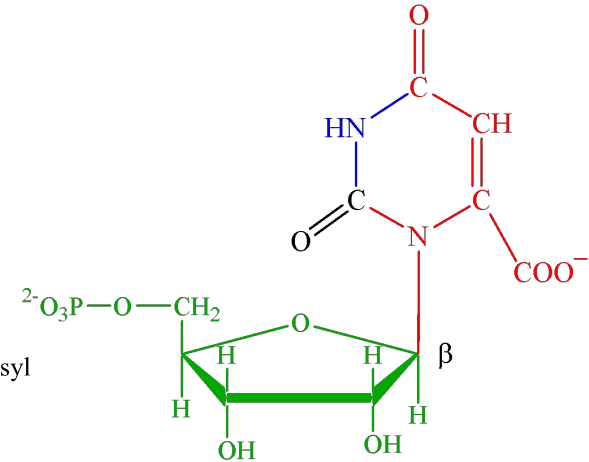
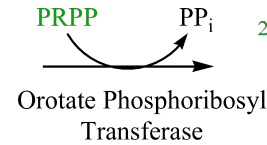
Carbamoyl Aspartate



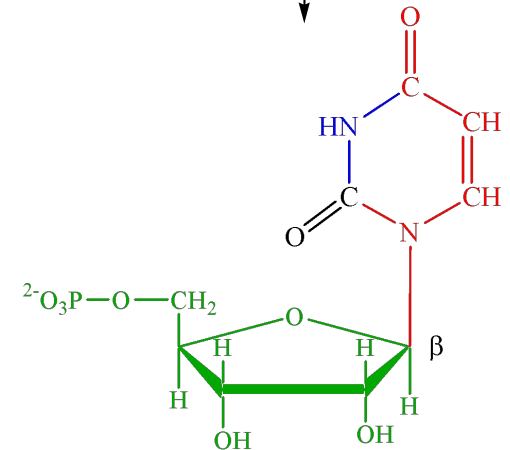
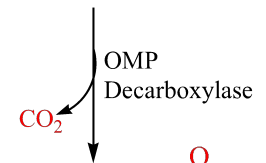
Dihydroorotate



Orotate



Orotidine-5'-monophosphate (OMP)



Uridine Monophosphate (UMP)

UMP $\rightleftharpoons$ UDP and UTP

- Nucleoside monophosphate kinase:



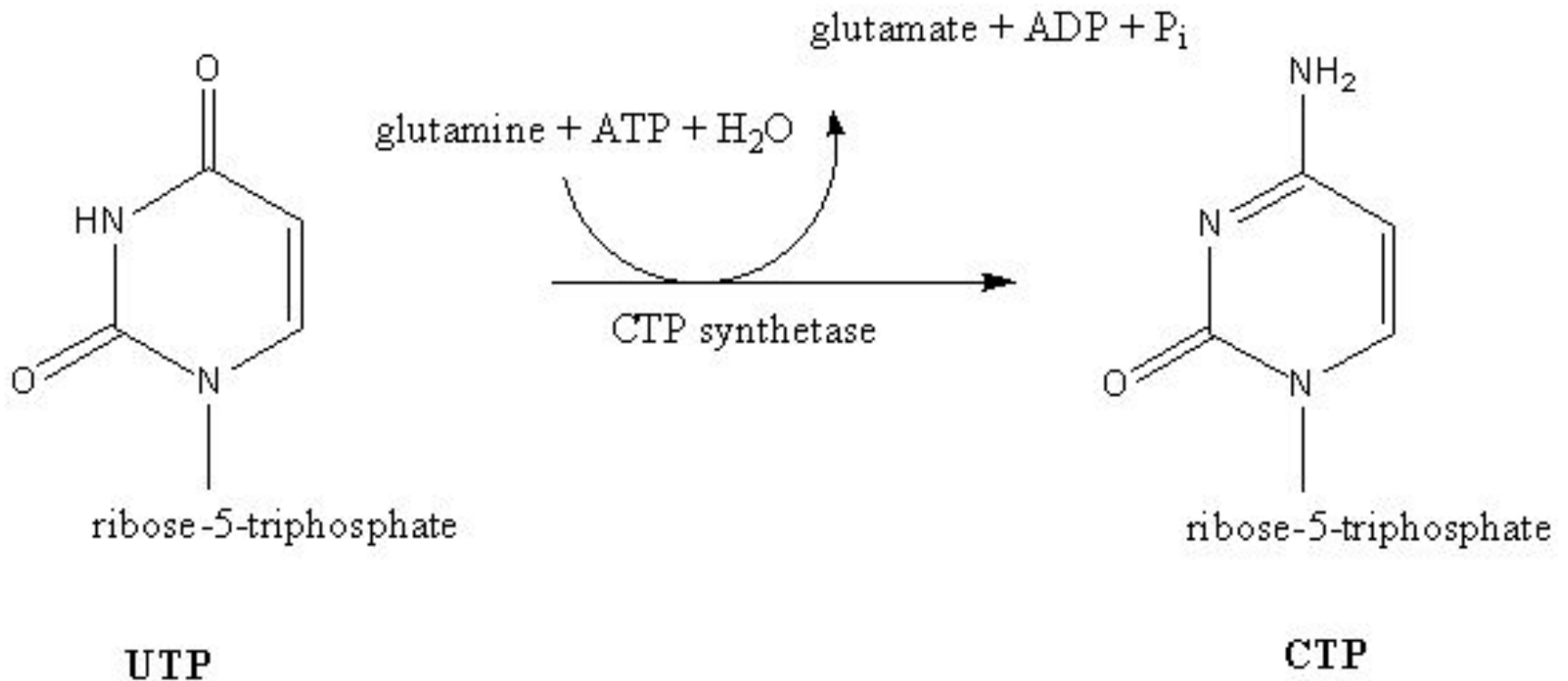
- Nucleoside diphosphate kinase:



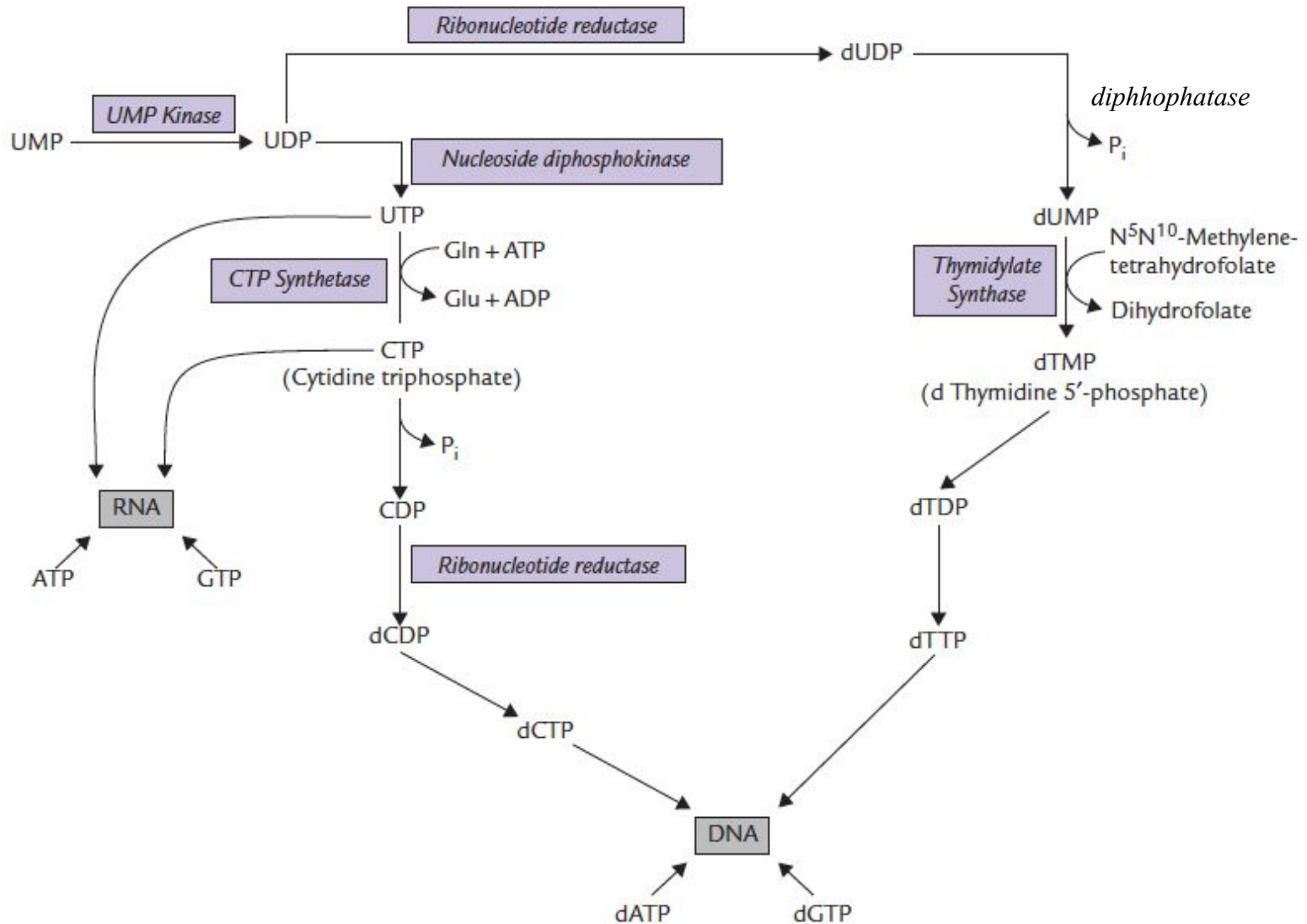
- UTP is used for formation of CTP, dCTP and dTTP
- UTP is needed for RNA production, but dUTP not needed for DNA
- If dUTP produced excessively, would cause substitution errors (dUTP for dTTP)

UTP $\square$ CTP

- CTP formed from UTP via CTP Synthetase driven by ATP hydrolysis
- Glutamine provides amide nitrogen for C4 in animals

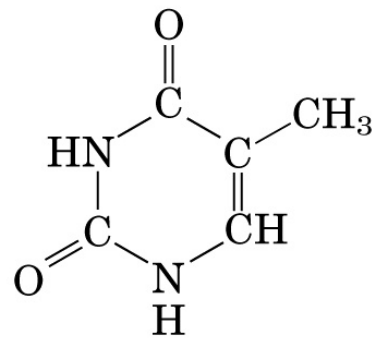


UDP → dCTP and dTTP

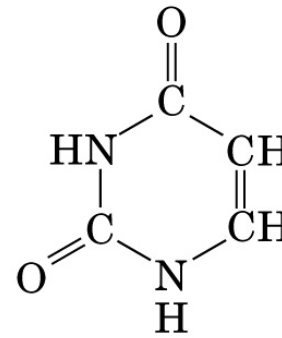


Thymine Nucleotide Formation

- Formed by methylating deoxyuridine monophosphate (dUMP)
- dUTP hydrolyzed by dUTPase (diphosphatase) $\rightarrow$ to dUMP
- dUMP $\rightarrow$ methylated at C5 to form dTMP $\rightarrow$ rephosphorylate to form dTTP



Thymine
(DNA)



Uracil
(RNA)

Tetrahydrofolate (THF)

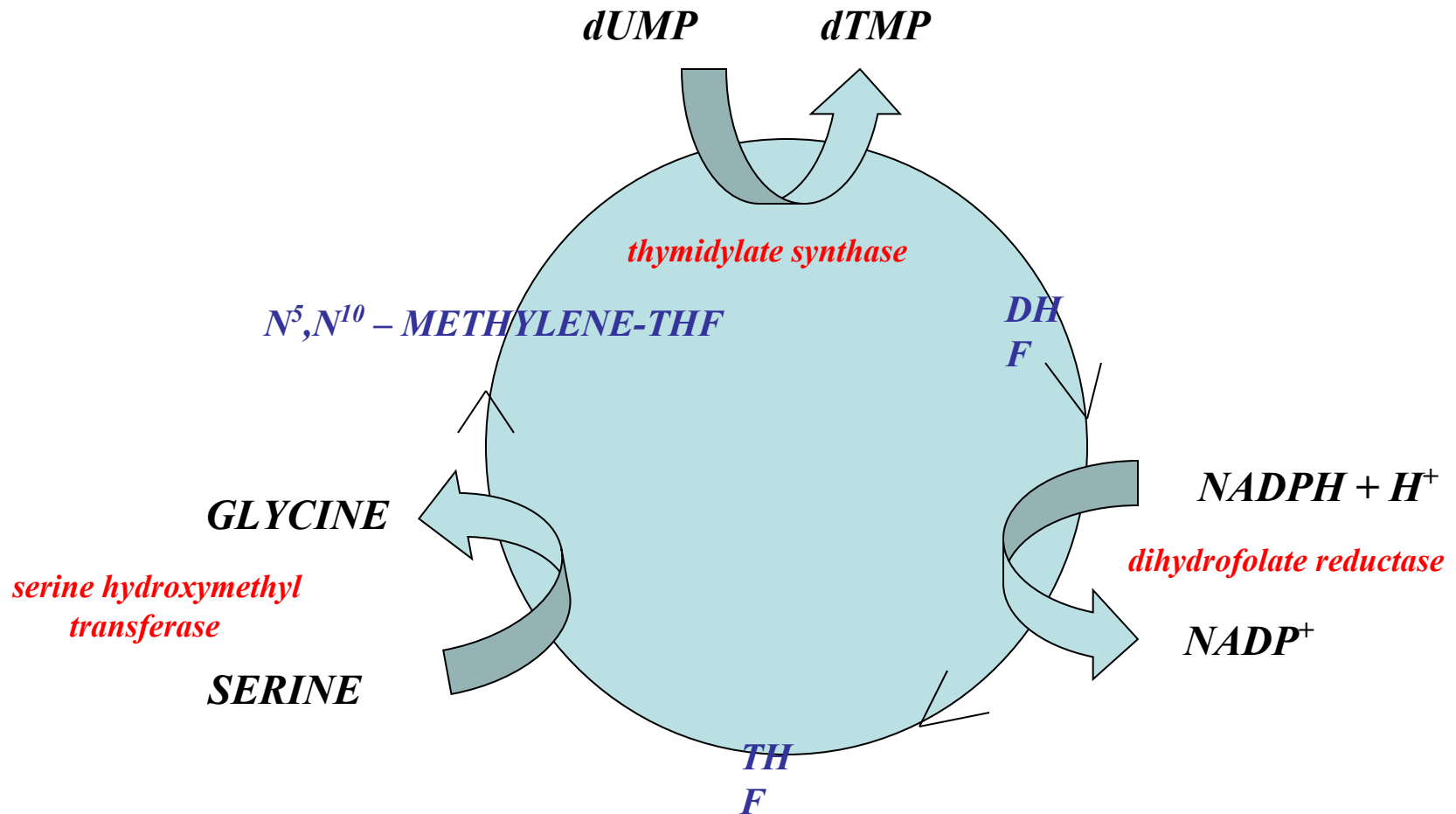
- Methylation of dUMP catalyzed by thymidylate synthase
 - Cofactor: N⁵,N¹⁰-methylene THF
 - Oxidized to dihydrofolate

- THF Regeneration:

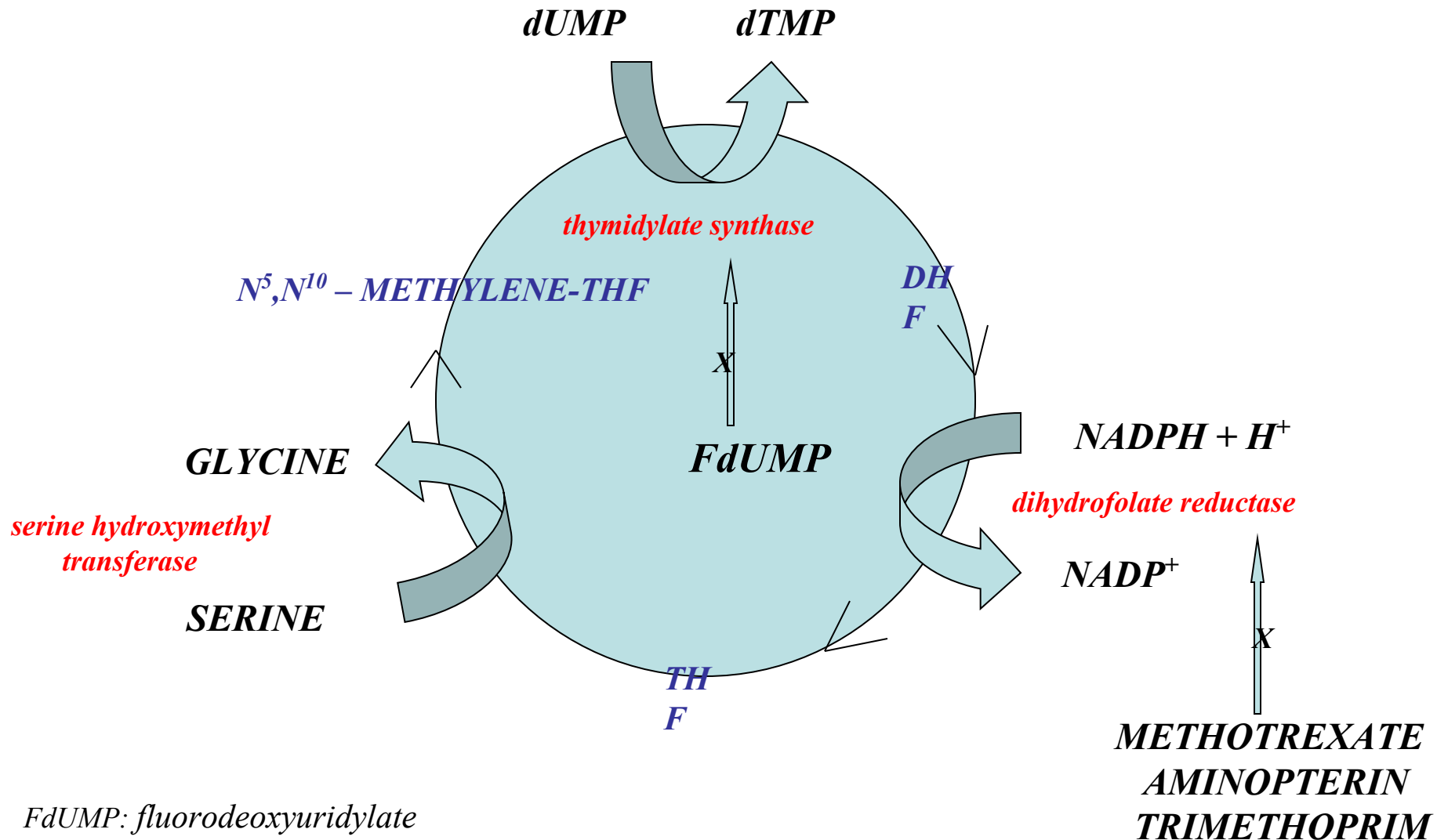
DHF + NADPH + H⁺ $\rightleftharpoons$ THF + NADP⁺ (enzyme: dihydrofolate reductase)

THF + Serine $\rightleftharpoons$ N⁵,N¹⁰-methylene-THF + Glycine
(enzyme: serine hydroxymethyl transferase)

REGENERATION OF N^5, N^{10} METHYLENETETRAHYDROFOLATE



INHIBITORS OF N^5,N^{10} METHYLENETETRAHYDROFOLATE REGENERATION



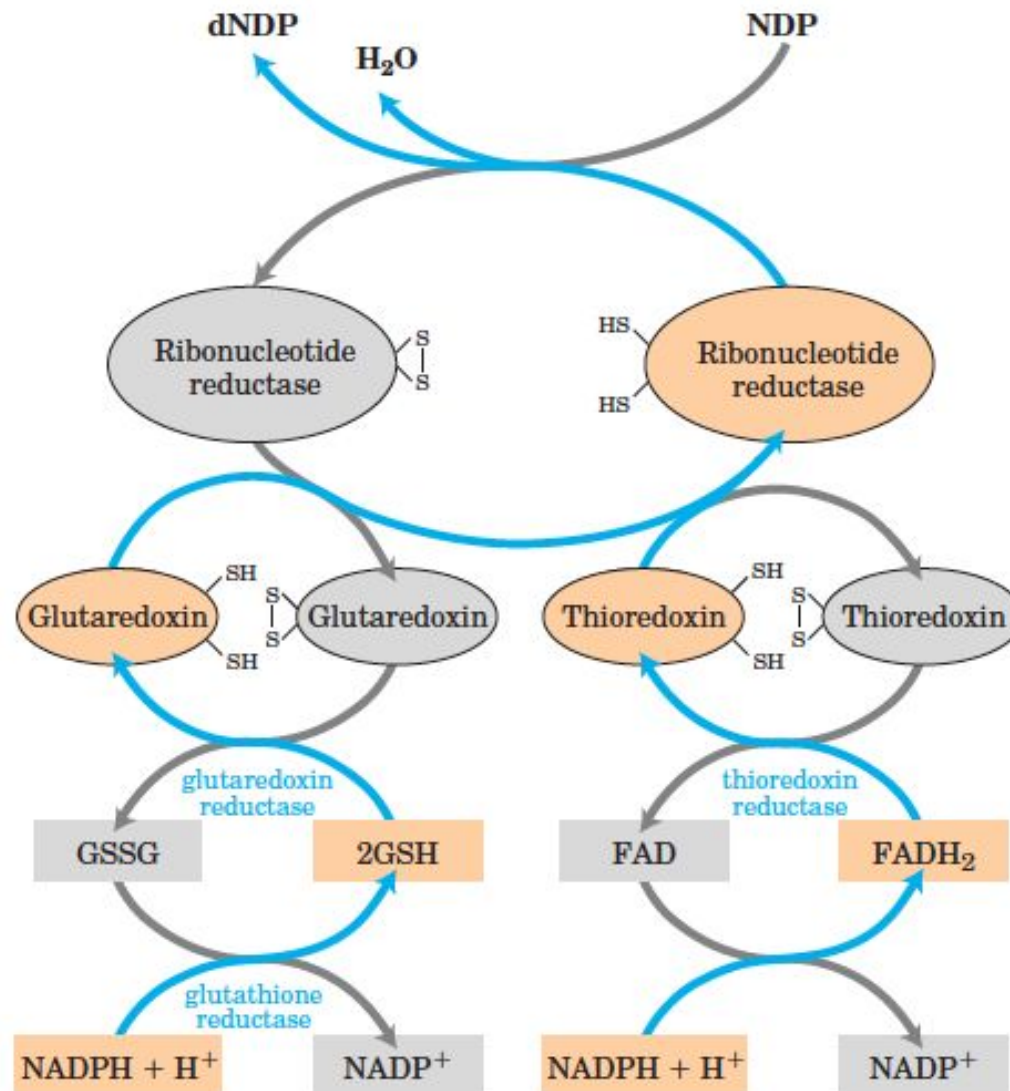
Anti-Folate Drugs

- Cancer cells consume dTMP quickly for DNA replication
 - Anti-folate drugs interfere with thymidylate synthase rxn to decrease dTMP production
 - Fluorodeoxyuridylate (irreversible inhibitor) – affects rapidly growing normal cells (hair follicles, bone marrow, immune system, intestinal mucosa)
- Dihydrofolate reductase step can be stopped competitively (DHF analogs)
 - Anti-Folates: Aminopterin, methotrexate, trimethoprim

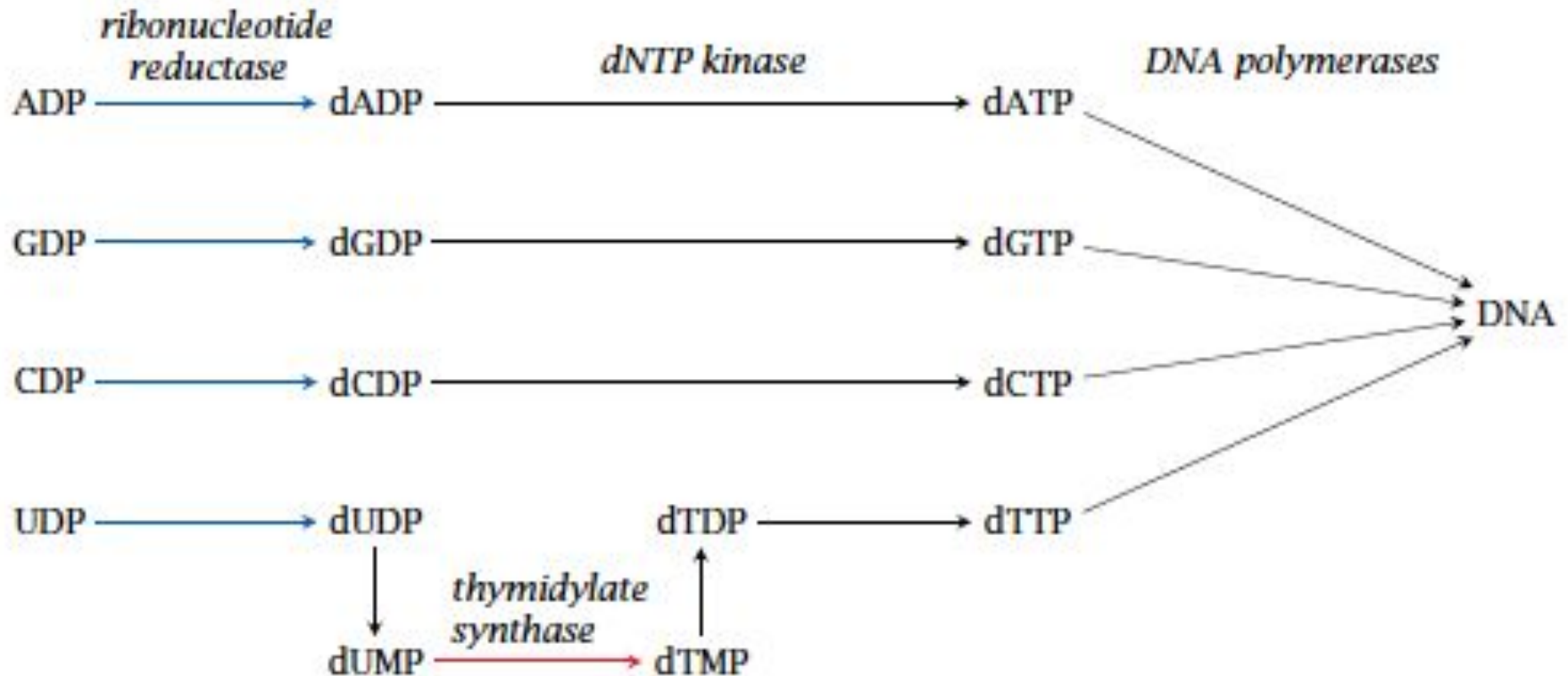
Deoxyribonucleotide Formation

- Deoxyribonucleotides are synthesized from corresponding ribonucleotides
- Reduction of 2' carbon done via a free radical mechanism
- Catalyzed by “Ribonucleotide Reductases”
 - *E. coli* RNR reduces ribonucleoside diphosphates (NDPs) to deoxyribonucleoside diphosphates (dNDPs)

Reduction of ribonucleotides to deoxyribonucleotides by ribonucleotide reductase.

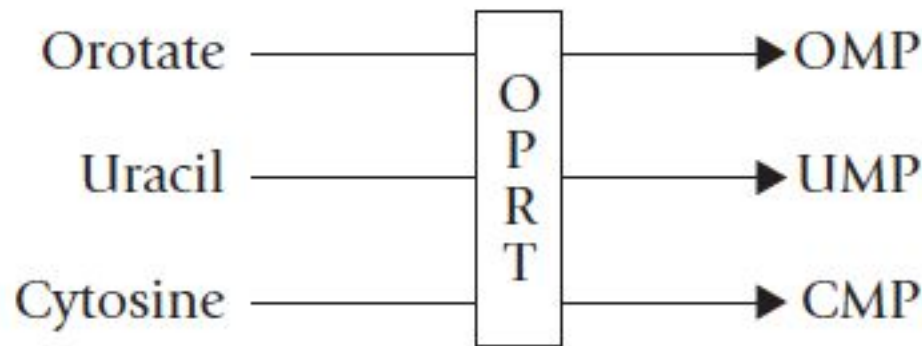


Synthesis of deoxyribonucleotides



Salvage pathway of Pyrimidine Synthesis

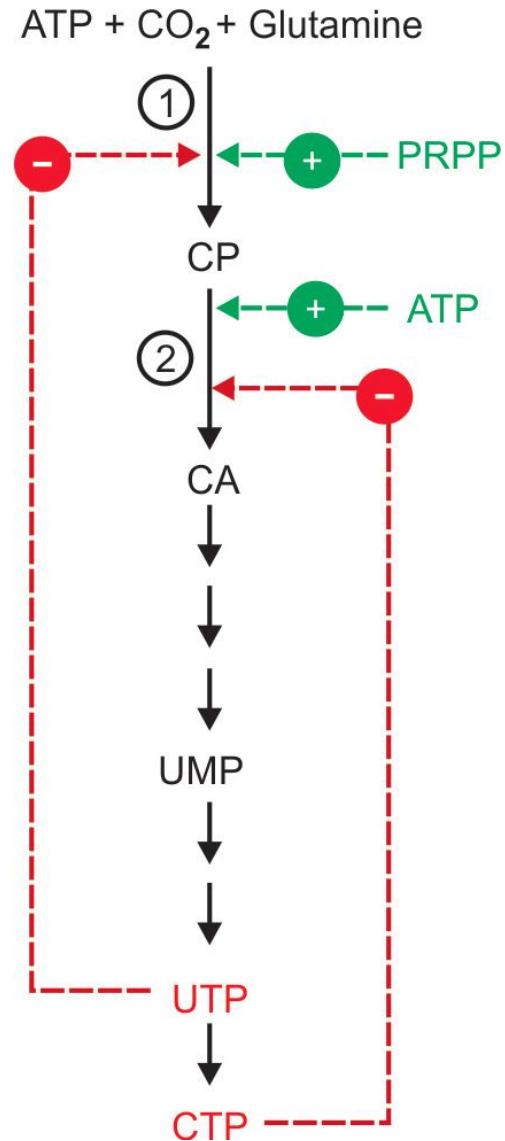
- Catalyzed by Orotate phosphoribosyl transferase (OPRT):
 - Catalyzes phosphoribosylation of orotate
 - Also salvages other pyrimidine bases, such as uracil and cytosine, by converting them to their corresponding nucleotides.



Regulatory Control of Pyrimidine Synthesis

- Differs between bacteria and animals
 - Bacteria – regulation at Aspartate Transcarbamoylase (ATCase) rxn
 - **Animals** – regulation at carbamoyl phosphate synthetase II & OMP Decarboxylase
 - UDP and UTP inhibit carbamoyl phosphate synthetase II; ATP and PRPP activate it
 - UMP and CMP competitively inhibit OMP Decarboxylase
- * Note: Purine synthesis inhibited by ADP and GDP at ribose phosphate pyrophosphokinase step, controlling level of PRPP □ also regulates pyrimidines synthesis.

Regulatory Control of Pyrimidine Synthesis



CP: Carbamoyl phosphate; CA: Carbamoyl aspartate

Orotic Aciduria

- Increased excretion of orotic acid in urine
- Normal - 1.4 mg/day; not toxic.
- An autosomal recessive disease.
- Caused by defect in enzyme activities (UMP synthase - bifunctional enzyme) of last two steps of pyrimidine synthesis i.e., OPRT and OMP decarboxylase.
- Type I – defect in both enzymic activities; Type II – defect in OMP decarboxylase activity.
- Excessive urinary level leads to its crystallization and consequent obstruction to urine flow
- Symptoms: retarded growth; severe anemia (megaloblastic), urinary tract obstruction
- Only known inherited defect in this pathway (all others would be lethal to fetus)
- Treat with uridine/cytidine; cannot be cured by Vit B₁₂ or Folic acid supplements

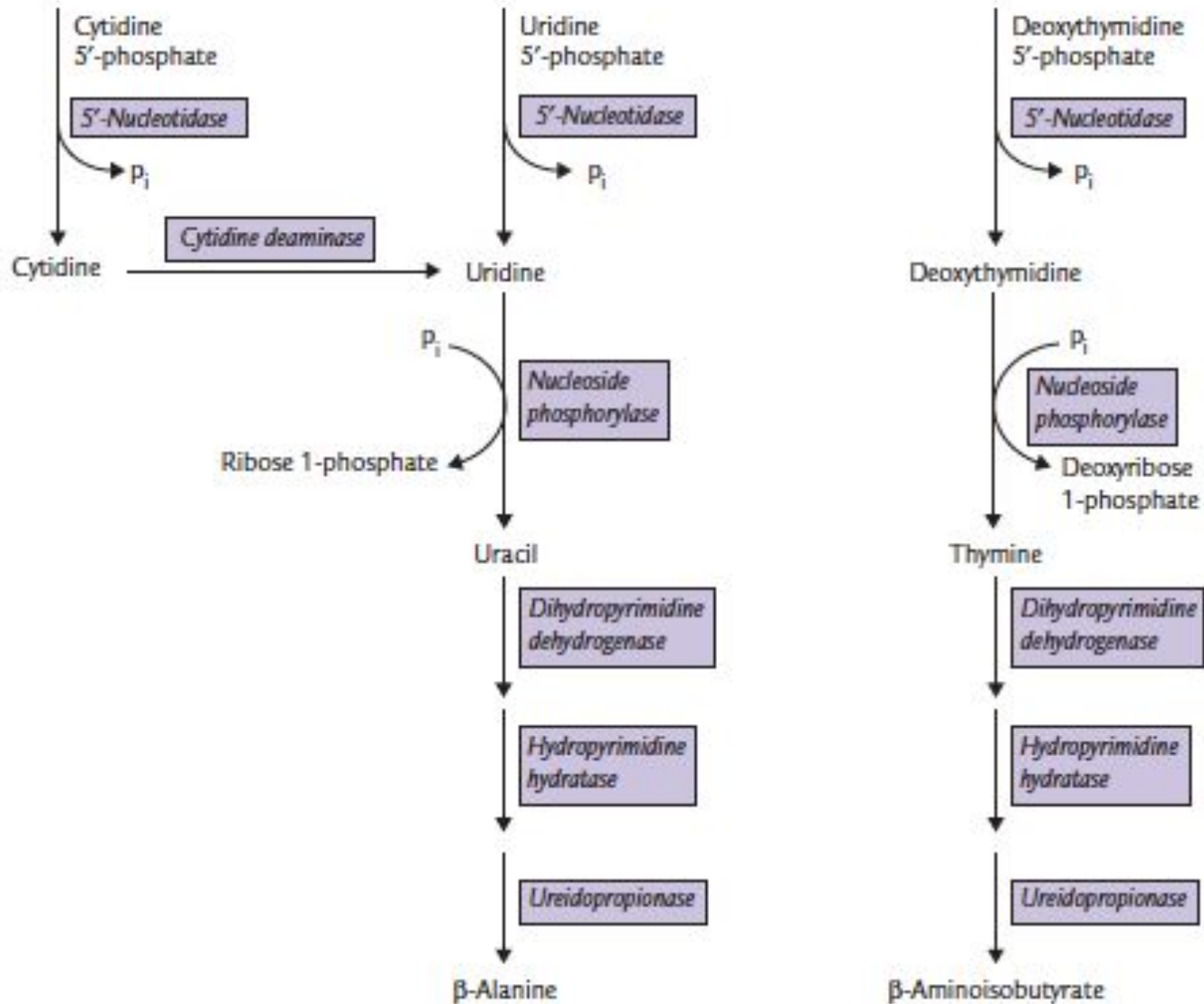
Secondary Orotic Aciduria: Rey's Syndrome

- Deficiency of ornithine transcarbamoylase, a urea cycle enzyme.
 - Leads to accumulation of carbamoyl phosphate, which is diverted for enhanced synthesis of orotic acid.

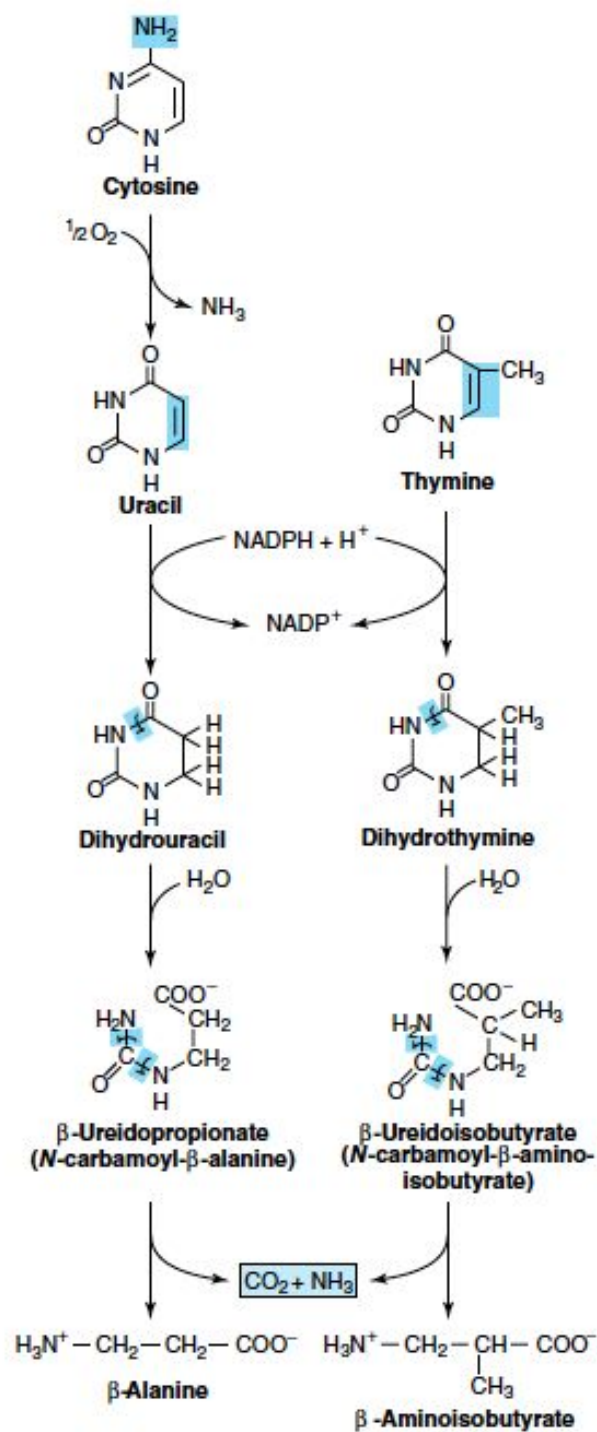
Degradation of Pyrimidines

- CMP and UMP degraded to bases similarly to purines
 - Dephosphorylation
 - Deamination
 - Glycosidic bond cleavage
- Uracil and thymine thus formed undergoes a series of reactions (ring splitting, reduction and hydrolytic cleavage in liver), forming β -alanine and β -aminoisobutyrate respectively.
 - Both compounds are transaminated and eventually metabolized to carbon dioxide although the exact pathway is not known.
 - β -alanine $\rightarrow$ malonate $\rightarrow$ malonyl-Co A $\rightarrow$ fatty acid synthesis.
 - β -aminoisobutyrate $\rightarrow$ methylmalonate semialdehyde $\rightarrow$ propionyl-Co A $\rightarrow$ methyl malonyl CoA $\rightarrow$ Succinyl CoA $\rightarrow$ TCA cycle
 - β -aminoisobutyrate induce fatty acid oxidation (anti-obesity factor) and insulin sensitivity.

Catabolism of pyrimidines.



Catabolism of pyrimidines.



Antagonists of nucleotide synthesis

- Glutamine antagonists, e.g. azaserine, antineoplastic drug.
 - Inhibit those steps in purine and pyrimidine metabolism in which glutamine donates a nitrogen.
- Structural analogues of bases or nucleosides: (see the next slide)
 - Act by inhibiting individual reactions in nucleotide metabolism or
 - Incorporate into DNA or RNA and make them functionally inactive.
 - Arrests cell division, hence useful in treatment of cancers.
 - e.g.: 5-fluorouracil (a uracil analogue); Azidothymidine (a nucleoside analog)
- Antifolates: (methotrexate, aminopterin, and related compounds)
 - Interfere with the action of folate cofactors - widely used in the treatment of cancer.
 - Inhibit the enzyme DHFR activity, thus blocking the synthesis of tetrahydrofolate
- 6-Mercaptopurine inhibits conversion of IMP to AMP and GMP

SYNTHETIC ANALOGUES OF NUCLEOTIDES OR ANTIMETABOLITES

- Prepared either by altering the heterocyclic ring or sugar moiety.
- Have therapeutic applications in medicine.
- Used in chemotherapy to control cancer or infections.
- Examples:
 - Azidothymidine (AZT; a Nucleoside analog): A structural analogue of thymidine used in the treatment of AIDS.
 - 5-iododeoxyuridine (a Nucleoside analog): A nucleoside analogue has antiviral activity and is used in the treatment of herpetic keratitis, an infection of the cornea by Herpes virus.
 - Cytarabine (arabinosyl cytosine; ara-c; a Nucleoside analog): In which arabinose replaces ribose, used in the chemotherapy of cancer and viral infections.
 - Allopurinol: A purine analogue, used in treatment of hyperuricemia and gout.

Read more...



THANK YOU...